8-3

1. A music group expects to sell a new compact disc (CD) at the rate $R(t) = 20,000e^{-0.12t}$ CDs per week, where t denotes the number of weeks since the CD was first released. To the nearest thousand, how many CDs are expected to be sold during the first 12 weeks after the release?

(A) 5,000

(B) 11,000

(C) 57,000

(D) 127,000 ✓

(E) 240,000

2. A home uses fuel oil at the rate $r(t) = 10 + 8\sin\left(\frac{t}{60}\right)$ gallons per day, where t is the number of days from the beginning of the heating season. To the nearest gallon, what is the total amount of fuel oil used from $t = 0$ to $t = 60$ days?

(A) 7 gal

(B) 14 gal

(C) 600 gal

(D) 821 gal ✓

(E) 1004 gal

3.

To help restore a beach, sand is being added to the beach at a rate of $s(t) = 65 + 24\sin(0.3t)$ tons per hour, where t is measured in hours since 5:00 A.M. How many tons of sand are added to the beach over the 3-hour period from 7:00 A.M. to 10:00 A.M.?

(A) 255.368 ✓

(B) 225.271

(C) 85.123

(D) 10.388

Answer A

This option is correct. The definite integral of the rate of change over an interval gives the net change over that interval. Since $s(t)$ is the rate of change at which sand is added to the beach, the definite integral of $s(t)$ on the interval $2 \le t \le 5$ gives the net change in the amount of sand on the beach over that time period. Since t is measured in hours since 5:00 A.M., the time 7:00 A.M. corresponds to $t = 2$ and 10:00 A.M. corresponds to $t = 5$.

$$\int_2^5 s(t)dt = \int_2^5 (65 + 24\sin(0.3t))dt = 255.368$$

8-3

4. Snow is falling at a rate of $r(t)=2e^{-0.1t}$ inches per hour, where t is the time in hours since the beginning of the snowfall. Which of the following expressions gives the amount of snow, in inches, that falls from time $t = 0$ to time $t = 5$ hours?

(A) $2e^{-0.5} - 2$

(B) $0.2 - 0.2e^{-0.5}$

(C) $4 - 4e^{-0.5}$

(D) $20 - 20e^{-0.5}$ ✓

Answer D

This option is correct. The definite integral of the rate of change over an interval gives the net change over that interval. Since $r(t)$ is the rate of change of the snowfall, the definite integral of $r(t)$ on the interval $0 \le t \le 5$ gives the change in the amount of snow that falls over that time period.

5. During a rainfall, the depth of water in a rain gauge increases at a rate modeled by $R(t) = 0.5 + t\cos\left(\frac{\pi t^3}{80}\right)$, where t is the time in hours since the start of the rainfall and $R(t)$ is measured in centimeters per hour. How much did the depth of water in the rain gauge increase from $t = 0$ to $t = 3$ hours?

(A) 1.233 cm

(B) 1.466 cm

(C) 1.966 cm

(D) 5.401 cm ✓

6. The rate at which motor oil is leaking from an automobile is modeled by the function L defined by $L(t) = 1 + \sin(t^2)$ for time $t \ge 0$. $L(t)$ is measured in liters per hour, and t is measured in hours. How much oil leaks out of the automobile during the first half hour?

(A) 1.998 liters

(B) 1.247 liters

(C) 0.969 liters

(D) 0.541 liters ✓

(E) 0.531 liters

8-3

7. At time $t = 0$ minutes, a tank contains 48 gallons of water. For $0 \le t \le 4$ minutes, water flows into the tank at a rate of $E(t) = 12t \sin\left(\frac{\pi}{4}t\right)$ gallons per minute, and water leaks out of the tank at a rate of $L(t) = \frac{6t^2}{t+2}$ gallons per minute. How many gallons of water are in the tank at time $t = 4$ minutes?

(A) 13.251

(B) 82.749

(C) 87.482

(D) 135.482

Answer B

This option is correct. The definite integral of the rate of change over an interval gives the net change over that interval. Since $E(t) - L(t)$ is the rate of change of the amount of water in the tank, the definite integral of $E(t) - L(t)$ on the interval $0 \le t \le 4$ gives the change in the amount of water over that time period. Adding the net change to the initial amount of water at $t = 0$ will give the amount of water at $t = 4$.

$$48 + \int_0^4 (E(t) - L(t))dt$$

$$= 48 + \int_0^4 \left(12t \sin\left(\frac{\pi}{4}t\right) - \frac{6t^2}{t+2}\right)dt = 48 + 34.749 = 82.749$$

The integration should be done with the calculator.

8-3

8. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

x	0	1	2	3	4	5	6
$f'(x)$	4	3.5	2	0.8	1.7	5.8	7

The function f satisfies $f(0) = 20$. The first derivative of f satisfies the inequality $0 \leq f'(x) \leq 7$ for all x in the closed interval $[0, 6]$. Selected values of f' are shown in the table above. The function f has a continuous second derivative for all real numbers.

(a) Use a midpoint Riemann sum with three subintervals of equal length indicated by the data in the table to approximate the value of $f(6)$.

(b) Determine whether the actual value of $f(6)$ could be 70. Explain your reasoning.

(c) Evaluate $\int_2^4 f''(x)dx$.

(d) Find $\lim_{x \to 0} \frac{f(x) - 20e^x}{0.5f(x) - 10}$.

Part A

1 point is earned for: midpoint sum

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_0^6 f'(x)\,dx \approx 2 \cdot 3.5 + 2 \cdot 0.8 + 2 \cdot 5.8 = 20.2$$

$$f(6) - f(0) = \int_0^6 f'(x)\,dx$$

$$f(6) = f(0) + \int_0^6 f'(x)\,dx \approx 20 + 20.2 = 40.2$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for: midpoint sum

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

8-3

$$\int_0^6 f'(x)\,dx \approx 2 \cdot 3.5 + 2 \cdot 0.8 + 2 \cdot 5.8 = 20.2$$

$$f(6) - f(0) = \int_0^6 f'(x)\,dx$$

$$f(6) = f(0) + \int_0^6 f'(x)\,dx \approx 20 + 20.2 = 40.2$$

Part B

1 point is earned for: integral bound

1 point is earned for: answer with reasoning

Since $f'(x) \le 7$, $\int_0^6 f'(x)\,dx \le 6 \cdot 7 = 42.$

$$f(6) - f(0) \le 42 \Rightarrow f(6) \le 20 + 42 = 62$$

Therefore, the actual value of $f(6)$ could not be 70.

0	1	2

The student response earns all of the following points:

1 point is earned for: integral bound

1 point is earned for: answer with reasoning

Since $f'(x) \le 7$, $\int_0^6 f'(x)\,dx \le 6 \cdot 7 = 42.$

$$f(6) - f(0) \le 42 \Rightarrow f(6) \le 20 + 42 = 62$$

Therefore, the actual value of $f(6)$ could not be 70.

Part C

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_2^4 f''(x)dx = f'(4) - f'(2) = 1.7 - 2 = -0.3$$

8-3

0	1	2

The student response earns all of the following points:

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_2^4 f''(x)dx = f'(4) - f'(2) = 1.7 - 2 = -0.3$$

Part D

1 point is earned for: L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x\to 0}(f(x) - 20e^x) = 0$$
$$\lim_{x\to 0}(0.5f(x) - 10) = 0$$

Using L'Hospital's Rule,

$$\lim_{x\to 0}\frac{f(x)-20e^x}{0.5f(x)-10} = \lim_{x\to 0}\frac{f'(x)-20e^x}{0.5f'(x)} = \frac{4-20}{0.5(4)} = -8$$

0	1	2

The student response earns all of the following points:

1 point is earned for: L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x\to 0}(f(x) - 20e^x) = 0$$
$$\lim_{x\to 0}(0.5f(x) - 10) = 0$$

Using L'Hospital's Rule,

$$\lim_{x\to 0}\frac{f(x)-20e^x}{0.5f(x)-10} = \lim_{x\to 0}\frac{f'(x)-20e^x}{0.5f'(x)} = \frac{4-20}{0.5(4)} = -8$$

8-3

9. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

x	0	1	2	3	4	5	6
$f'(x)$	4	3.5	2	0.8	1.7	5.8	7

The function f satisfies $f(0) = 20$. The first derivative of f satisfies the inequality $0 \leq f'(x) \leq 7$ for all x in the closed interval $[0,\ 6]$. Selected values of f' are shown in the table above. The function f has a continuous second derivative for all real numbers.

(a) Use a midpoint Riemann sum with three subintervals of equal length indicated by the data in the table to approximate the value of $f(6)$.

(b) Determine whether the actual value of $f(6)$ could be 70. Explain your reasoning.

(c) Evaluate $\int_2^4 f''(x)dx$.

(d) Find $\lim_{x \to 0} \frac{f(x) - 20e^x}{0.5f(x) - 10}$.

Part A

1 point is earned for: midpoint sum

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_0^6 f'(x)\,dx \approx 2 \cdot 3.5 + 2 \cdot 0.8 + 2 \cdot 5.8 = 20.2$$

$$f(6) - f(0) = \int_0^6 f'(x)\,dx$$

$$f(6) = f(0) + \int_0^6 f'(x)\,dx \approx 20 + 20.2 = 40.2$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for: midpoint sum

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

8-3

$$\int_0^6 f'(x)\,dx \approx 2\cdot 3.5 + 2\cdot 0.8 + 2\cdot 5.8 = 20.2$$

$$f(6) - f(0) = \int_0^6 f'(x)\,dx$$

$$f(6) = f(0) + \int_0^6 f'(x)\,dx \approx 20 + 20.2 = 40.2$$

Part B

1 point is earned for: integral bound

1 point is earned for: answer with reasoning

Since $f'(x) \le 7$, $\int_0^6 f'(x)\,dx \le 6\cdot 7 = 42$.

$$f(6) - f(0) \le 42 \Rightarrow f(6) \le 20 + 42 = 62$$

Therefore, the actual value of $f(6)$ could not be 70.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for: integral bound

1 point is earned for: answer with reasoning

Since $f'(x) \le 7$, $\int_0^6 f'(x)\,dx \le 6\cdot 7 = 42$.

$$f(6) - f(0) \le 42 \Rightarrow f(6) \le 20 + 42 = 62$$

Therefore, the actual value of $f(6)$ could not be 70.

Part C

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_2^4 f''(x)dx = f'(4) - f'(2) = 1.7 - 2 = -0.3$$

8-3

0	1	2

The student response earns all of the following points:

1 point is earned for: Fundamental Theorem of Calculus

1 point is earned for: answer

$$\int_{2}^{4} f''(x)dx = f'(4) - f'(2) = 1.7 - 2 = -0.3$$

Part D

1 point is earned for: L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x\to 0}(f(x) - 20e^{x}) = 0$$
$$\lim_{x\to 0}(0.5f(x) - 10) = 0$$

Using L'Hospital's Rule,

$$\lim_{x\to 0}\frac{f(x)-20e^{x}}{0.5f(x)-10} = \lim_{x\to 0}\frac{f'(x)-20e^{x}}{0.5f'(x)} = \frac{4-20}{0.5(4)} = -8$$

0	1	2

The student response earns all of the following points:

1 point is earned for: L'Hospital's Rule

1 point is earned for: answer

$$\lim_{x\to 0}(f(x) - 20e^{x}) = 0$$
$$\lim_{x\to 0}(0.5f(x) - 10) = 0$$

Using L'Hospital's Rule,

$$\lim_{x\to 0}\frac{f(x)-20e^{x}}{0.5f(x)-10} = \lim_{x\to 0}\frac{f'(x)-20e^{x}}{0.5f'(x)} = \frac{4-20}{0.5(4)} = -8$$

8-3

10.

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W, where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t = 0$, the temperature of the water is 55°F. The water is heated for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

(a) Use the data in the table to estimate $W'(12)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.

(b) Use the data in the table to evaluate $\int_0^{20} W'(t)dt$. Using correct units, interpret the meaning of $\int_0^{20} W'(t)dt$ in the context of this problem.

(c) For $0 \le t \le 20$, the average temperature of the water in the tub is $\frac{1}{20}\int_0^{20} W(t)dt$. Use a left Riemann sum with the four subintervals indicated by the data in the table to approximate $\frac{1}{20}\int_0^{20} W(t)dt$. Does this approximation overestimate or underestimate the average temperature of the water over these 20 minutes? Explain your reasoning.

(d) For $20 \le t \le 25$, the function W that models the water temperature has first derivative given by $W'(t) = 0.4\sqrt{t}\cos(0.06t)$. Based on the model, what is the temperature of the water at time $t = 25$?

Part A

1 point is earned for estimation

1 point is earned for interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

= 1.017 (or 1.016)

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.

0	1	2

The student earns all of the following points:

1 point is earned for estimation

1 point is earned for interpretation with units

8-3

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

= 1.017 (or 1.016)

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.

Part B

1 point is earned for the value

1 point is earned for the interpretation with units

$$\int_0^{20} W'(t)dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.

0	1	2 ✓

The student earns all of the following points:

1 point is earned for the value

1 point is earned for the interpretation with units

$$\int_0^{20} W'(t)dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.

Part C

1 point is earned for left Riemann sum

1 point is earned for approximation

1 point is earned for underestimating with reason

8-3

$$\frac{1}{20}\int_0^{20} W(t)dt \approx \frac{1}{20}(4\cdot W(0)+5\cdot W(4)+6\cdot W(9)+5\cdot W(15))$$
$$=\frac{1}{20}(4\cdot 55.0+5\cdot 57.1+6\cdot 61.8+5\cdot 67.9)$$
$$=\frac{1}{20}\cdot 1215.8=60.79$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing

0	1	2	3

The student earns all of the following points:

1 point is earned for left Riemann sum

1 point is earned for approximation

1 point is earned for underestimating with reason

$$\frac{1}{20}\int_0^{20} W(t)dt \approx \frac{1}{20}(4\cdot W(0)+5\cdot W(4)+6\cdot W(9)+5\cdot W(15))$$
$$=\frac{1}{20}(4\cdot 55.0+5\cdot 57.1+6\cdot 61.8+5\cdot 67.9)$$
$$=\frac{1}{20}\cdot 1215.8=60.79$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing

Part D

1 point is earned for the integral

1 point is earned for the answer

$$W(25)=71.0+\int_{20}^{25} W'(t)dt$$
$$=71.0+2.043155=73.043$$

0	1	2

8-3

The student earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned} W(25) &= 71.0 + \int_{20}^{25} W'(t)dt \\ &= 71.0 + 2.043155 = 73.043 \end{aligned}$$

8-3

11.

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W, where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t = 0$, the temperature of the water is 55° F. The water is heated for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

Use the data in the table to evaluate $\int_0^{20} W'(t)dt$. Using correct units, interpret the meaning of $\int_0^{20} W'(t)dt$ in the context of this problem.

Part B

One point is earned for the value

One point is earned for the interpretation with units

$$\int_0^{20} W'(t)dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16 °F over the interval from $t = 0$ to $t = 20$ minutes.

0	1	2

The student earns all of the following points:

One point is earned for the value

One point is earned for the interpretation with units

$$\int_0^{20} W'(t)dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16 °F over the interval from $t = 0$ to $t = 20$ minutes.

8-3

12. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

t (minutes)	0	1	5	6	8
$g(t)$ (cubic feet per minute)	12.8	15.1	20.5	18.3	22.7

Grain is being added to a silo. At time $t = 0$, the silo is empty. The rate at which grain is being added is modeled by the differentiable function g, where $g(t)$ is measured in cubic feet per minute for $0 \le t \le 8$ minutes. Selected values of $g(t)$ are given in the table above.

(a) Using the data in the table, approximate $g'(3)$. Using correct units, interpret the meaning of $g'(3)$ in the context of the problem.

(b) Write an integral expression that represents the total amount of grain added to the silo from time $t = 0$ to time $t = 8$. Use a right Riemann sum with the four subintervals indicated by the data in the table to approximate the integral.

(c) The grain in the silo is spoiling at a rate modeled by $w(t) = 32 \cdot \sqrt{\sin\left(\frac{\pi t}{74}\right)}$, where $w(t)$ is measured in cubic feet per minute for $0 \le t \le 8$ minutes. Using the result from part (b), approximate the amount of unspoiled grain remaining in the silo at time $t = 8$.

(d) Based on the model in part (c), is the amount of unspoiled grain in the silo increasing or decreasing at time $t = 6$? Show the work that leads to your answer.

Part A

1 point is earned for: approximation

1 point is earned for: interpretation with units

$$g'(3) \approx \frac{g(5)-g(1)}{5-1} = \frac{20.5-15.1}{4} = 1.35$$

At time $t = 3$ minutes, the rate at which grain is being added to the silo is increasing at a rate of 1.35 cubic feet per minute per minute.

0	1	2

The student response earns all of the following points:

1 point is earned for: approximation

1 point is earned for: interpretation with units

$$g'(3) \approx \frac{g(5)-g(1)}{5-1} = \frac{20.5-15.1}{4} = 1.35$$

At time $t = 3$ minutes, the rate at which grain is being added to the silo is increasing at a rate of 1.35 cubic feet per minute per minute.

8-3

Part B

1 point is earned for: integral expression

1 point is earned for: right Riemann sum

1 point is earned for: approximation

The total amount of grain added to the silo from time $t = 0$ to time $t = 8$ is $\int_0^8 g(t)\,dt$ cubic feet.

$$\int_0^8 g(t)\,dt \approx g(1)\cdot(1-0) + g(5)\cdot(5-1) + g(6)\cdot(6-5) + g(8)\cdot(8-6)$$
$$= 15.1\cdot 1 + 20.5\cdot 4 + 18.3\cdot 1 + 22.7\cdot 2 = 160.8$$

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for: integral expression

1 point is earned for: right Riemann sum

1 point is earned for: approximation

The total amount of grain added to the silo from time $t = 0$ to time $t = 8$ is $\int_0^8 g(t)\,dt$ cubic feet.

$$\int_0^8 g(t)\,dt \approx g(1)\cdot(1-0) + g(5)\cdot(5-1) + g(6)\cdot(6-5) + g(8)\cdot(8-6)$$
$$= 15.1\cdot 1 + 20.5\cdot 4 + 18.3\cdot 1 + 22.7\cdot 2 = 160.8$$

Part C

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^8 w(t)\,dt = 99.051497$$

The approximate amount of unspoiled grain remaining in the silo at time $t = 8$ is $160.8 - \int_0^8 w(t)\,dt = 61.749$ (or 61.748) cubic feet.

8-3

0	1	2

The student response earns all of the following points:

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^8 w(t)\ dt = 99.051497$$

The approximate amount of unspoiled grain remaining in the silo at time $t = 8$ is $160.8 - \int_0^8 w(t)\ dt = 61.749$ (or 61.748) cubic feet.

Part D

1 point is earned for: considers $g(6) - w(6)$

1 point is earned for: answer

$g(6) - w(6) = 18.3 - 16.063173 = 2.236827 > 0$

Because $g\,(6) - w\,(6) > 0$, the amount of unspoiled grain is increasing at time $t = 6$.

0	1	2

The student response earns all of the following points:

1 point is earned for: considers $g(6) - w(6)$

1 point is earned for: answer

$g(6) - w(6) = 18.3 - 16.063173 = 2.236827 > 0$

Because $g\,(6) - w\,(6) > 0$, the amount of unspoiled grain is increasing at time $t = 6$.

8-3

13. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

t (minutes)	0	1	5	6	8
$g(t)$ (cubic feet per minute)	12.8	15.1	20.5	18.3	22.7

Grain is being added to a silo. At time $t = 0$, the silo is empty. The rate at which grain is being added is modeled by the differentiable function g, where $g(t)$ is measured in cubic feet per minute for $0 \leq t \leq 8$ minutes. Selected values of $g(t)$ are given in the table above.

(a) Using the data in the table, approximate $g'(3)$. Using correct units, interpret the meaning of $g'(3)$ in the context of the problem.

(b) Write an integral expression that represents the total amount of grain added to the silo from time $t = 0$ to time $t = 8$. Use a right Riemann sum with the four subintervals indicated by the data in the table to approximate the integral.

(c) The grain in the silo is spoiling at a rate modeled by $w(t) = 32 \cdot \sqrt{\sin\left(\frac{\pi t}{74}\right)}$, where $w(t)$ is measured in cubic feet per minute for $0 \leq t \leq 8$ minutes. Using the result from part (b), approximate the amount of unspoiled grain remaining in the silo at time $t = 8$.

(d) Based on the model in part (c), is the amount of unspoiled grain in the silo increasing or decreasing at time $t = 6$? Show the work that leads to your answer.

Part A

1 point is earned for: approximation

1 point is earned for: interpretation with units

$$g'(3) \approx \frac{g(5)-g(1)}{5-1} = \frac{20.5-15.1}{4} = 1.35$$

At time $t = 3$ minutes, the rate at which grain is being added to the silo is increasing at a rate of 1.35 cubic feet per minute per minute.

0	1	2

The student response earns all of the following points:

1 point is earned for: approximation

1 point is earned for: interpretation with units

$$g'(3) \approx \frac{g(5)-g(1)}{5-1} = \frac{20.5-15.1}{4} = 1.35$$

8-3

At time $t = 3$ minutes, the rate at which grain is being added to the silo is increasing at a rate of 1.35 cubic feet per minute per minute.

Part B

1 point is earned for: integral expression

1 point is earned for: right Riemann sum

1 point is earned for: approximation

The total amount of grain added to the silo from time $t = 0$ to time $t = 8$ is $\int_0^8 g(t)\,dt$ cubic feet.

$$\begin{aligned}\int_0^8 g(t)\,dt &\approx g(1)\cdot(1-0) + g(5)\cdot(5-1) + g(6)\cdot(6-5) + g(8)\cdot(8-6)\\ &= 15.1\cdot 1 + 20.5\cdot 4 + 18.3\cdot 1 + 22.7\cdot 2 = 160.8\end{aligned}$$

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for: integral expression

1 point is earned for: right Riemann sum

1 point is earned for: approximation

The total amount of grain added to the silo from time $t = 0$ to time $t = 8$ is $\int_0^8 g(t)\,dt$ cubic feet.

$$\begin{aligned}\int_0^8 g(t)\,dt &\approx g(1)\cdot(1-0) + g(5)\cdot(5-1) + g(6)\cdot(6-5) + g(8)\cdot(8-6)\\ &= 15.1\cdot 1 + 20.5\cdot 4 + 18.3\cdot 1 + 22.7\cdot 2 = 160.8\end{aligned}$$

Part C

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^8 w(t)\,dt = 99.051497$$

8-3

The approximate amount of unspoiled grain remaining in the silo at time $t = 8$ is $160.8 - \int_0^8 w(t)\, dt = 61.749$ (or 61.748) cubic feet.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^8 w(t)\, dt = 99.051497$$

The approximate amount of unspoiled grain remaining in the silo at time $t = 8$ is $160.8 - \int_0^8 w(t)\, dt = 61.749$ (or 61.748) cubic feet.

Part D

1 point is earned for: considers $g(6) - w(6)$

1 point is earned for: answer

$g(6) - w(6) = 18.3 - 16.063173 = 2.236827 > 0$

Because $g(6) - w(6) > 0$, the amount of unspoiled grain is increasing at time $t = 6$.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for: considers $g(6) - w(6)$

1 point is earned for: answer

$g(6) - w(6) = 18.3 - 16.063173 = 2.236827 > 0$

Because $g(6) - w(6) > 0$, the amount of unspoiled grain is increasing at time $t = 6$.

8-3

14. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

For $0 \leq t \leq 6$ seconds, a screen saver on a computer screen shows two circles that start as dots and expand outward.

(a) At the instant that the first circle has a radius of 9 centimeters, the radius is increasing at a rate of $\frac{3}{2}$ centimeters per second. Find the rate at which the area of the circle is changing at that instant. Indicate units of measure.

(b) The radius of the first circle is modeled by $w(t) = 12 - 12e^{-0.5t}$ for $0 \leq t \leq 6$, where $w(t)$ is measured in centimeters and t is measured in seconds. At what time t is the radius of the circle increasing at a rate of 3 centimeters per second?

(c) A model for the radius of the second circle is given by the function f for $0 \leq t \leq 6$, where $f(t)$ is measured in centimeters and t is measured in seconds. The rate of change of the radius of the second circle is given by $f'(t) = t^2 - 4t + 4$. Based on this model, by how many centimeters does the radius of the second circle increase from time $t = 0$ to $t = 3$?

Part A

1 point is earned for: $\frac{dA}{dt}$

1 point is earned for: answer

1 point is earned for: units

$$A = \pi r^2$$
$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$
$$\left.\frac{dA}{dt}\right|_{r=9} = 2\pi \cdot 9 \cdot \frac{3}{2} = 27\pi$$

When the radius is 9 centimeters, the area is changing at a rate of $27\pi\ \mathrm{cm}^2/\mathrm{sec}$.

0	1	2	3

The student response earns all of the following points:

1 point is earned for: $\frac{dA}{dt}$

1 point is earned for: answer

1 point is earned for: units

$$A = \pi r^2$$
$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$
$$\left.\frac{dA}{dt}\right|_{r=9} = 2\pi \cdot 9 \cdot \frac{3}{2} = 27\pi$$

8-3

When the radius is 9 centimeters, the area is changing at a rate of $27\pi\ \text{cm}^2/\text{sec}$.

Part B

1 point is earned for: $w'(t)$

1 point is earned for: sets $w'(t) = 3$

1 point is earned for: answer

$$w'(t) = (-12)(-0.5)e^{-0.5t} = 6e^{-0.5t}$$
$$6e^{-0.5t} = 3 \Rightarrow e^{-0.5t} = \frac{1}{2} \Rightarrow -0.5t = \ln\left(\frac{1}{2}\right) \Rightarrow t = 2\ln 2$$

The radius is increasing at a rate of 3 centimeters per second at time$t = 2\ln 2$seconds.

0	1	2	3

The student response earns all of the following points:

1 point is earned for: $w'(t)$

1 point is earned for: sets $w'(t) = 3$

1 point is earned for: answer

$$w'(t) = (-12)(-0.5)e^{-0.5t} = 6e^{-0.5t}$$
$$6e^{-0.5t} = 3 \Rightarrow e^{-0.5t} = \frac{1}{2} \Rightarrow -0.5t = \ln\left(\frac{1}{2}\right) \Rightarrow t = 2\ln 2$$

The radius is increasing at a rate of 3 centimeters per second at time $t = 2\ln 2$ seconds.

Part C

1 point is earned for: integral

1 point is earned for: antiderivative

1 point is earned for: answer

$$\begin{aligned}\int_0^3 \left(t^2 - 4t + 4\right) dt &= \left[\tfrac{1}{3}t^3 - 2t^2 + 4t\right]_{t=0}^{t=3} \\ &= \left(\tfrac{1}{3}\cdot 3^3 - 2\cdot 3^2 + 4\cdot 3\right) - \left(\tfrac{1}{3}\cdot 0^3 - 2\cdot 0^2 + 4\cdot 0\right) \\ &= 9 - 18 + 12 = 3\end{aligned}$$

The radius increases by 3 centimeters from time $t = 0$ to time $t = 3$ seconds.

8-3

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for: integral

1 point is earned for: antiderivative

1 point is earned for: answer

$$\int_0^3 \left(t^2 - 4t + 4\right) dt = \left[\tfrac{1}{3}t^3 - 2t^2 + 4t\right]_{t=0}^{t=3}$$
$$= \left(\tfrac{1}{3} \cdot 3^3 - 2 \cdot 3^2 + 4 \cdot 3\right) - \left(\tfrac{1}{3} \cdot 0^3 - 2 \cdot 0^2 + 4 \cdot 0\right)$$
$$= 9 - 18 + 12 = 3$$

The radius increases by 3 centimeters from time $t = 0$ to time $t = 3$ seconds.

8-3

15. The following are related to this scenario:

A store is having a 12-hour sale. The total number of shoppers who have entered the store t hours after the sale begins is modeled by the function S defined by $S(t)=0.5t^4-16t^3+144t^2$ for $0 \le t \le 12$. At time $t=0$, when the sale begins, there are no shoppers in the store.

The rate at which shoppers leave the store, measured in shoppers per hour, is modeled by the function L defined by $L(t) = -80 + \frac{4400}{t^2-14t+55}$ for $0 \le t \le 12$. According to the model, how many shoppers are in the store at the end of the sale (time $t = 12$)? Give your answers to the nearest whole number.

Part C

The response can earn up to 3 points:

1 point: For correct integral

1 point: For use of $S\,(12)$

1 point: For the correct answer

$$S(12) - \int_0^{12} L(t)dt = 195.701684$$

Therefore there are approximately 196 shoppers in the store at the end of the sale.

0	1	2	3 ✓

The response earns all three of the following points:

1 point: For correct integral

1 point: For use of $S\,(12)$

1 point: For the correct answer

$$S(12) - \int_0^{12} L(t)dt = 195.701684$$

Therefore there are approximately 196 shoppers in the store at the end of the sale.

8-3

16. The rate at which raw sewage enters a treatment tank is given by $E(t) = 850 + 715\cos\left(\frac{\pi t^2}{9}\right)$ gallons per hour for $0 \le t \le 4$ hours. Treated sewage is removed from the tank at the constant rate of 645 gallons per hour. The treatment tank is empty at time $t = 0$.

How many gallons of sewage enter the treatment tank during the time interval $0 \le t \le 4$ Round your answer to the nearest gallon.

Part A

1 point is earned for integral

1 point is earned for the answer

$$\int_0^4 E(t)dt = 3981 gallons$$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

$$\int_0^4 E(t)dt = 3981 gallons$$

17. A cup of tea is cooling in a room that has a constant temperature of 70 degrees Fahrenheit $({}^\circ F)$. If the initial temperature of the tea, at time $t = 0$ minutes, is $200{}^\circ F$ and the temperature of the tea changes at the rate $R(t) = -6.89e^{-0.053t}$ degrees Fahrenheit per minute, what is the temperature, to the nearest degree, of the tea after 4 minutes?

(A) $175{}^\circ F$ ✓

(B) $130{}^\circ F$

(C) $95{}^\circ F$

(D) $70{}^\circ F$

(E) $45{}^\circ F$

8-3

18. A particle moves along a line so that its acceleration for $t \geq 0$ is given by $a(t) = \frac{t+3}{\sqrt{t^3+1}}$. If the particle's velocity at $t = 0$ is 5, what is the velocity of the particle at $t = 3$?

(A) 0.713

(B) 1.134

(C) 6.134

(D) 6.710

(E) 11.710

8-3

The penguin population on an island is modeled by a differentiable function P of time t, where $P(t)$ is the number of penguins and t is measured in years for $0 \le t \le 40$. There are 100,000 penguins on the island at time $t = 0$. The birth rate for the penguins on the island is modeled by

$B(t) = 1000e^{0.06t}$ penguins per year

and the death rate for the penguins on the island is modeled by

$D(t)=250e^{0.1t}$ penguins per year.

19. To the nearest whole number, what is the penguin population on the island at time $t = 40$?

Part B

The response can earn up to 2 points:

1 point: For correct limits

1 point: For the integrand

1 point: For the correct answer

$$P(40) = 100000 + \int_0^{40} (B(t) - D(t))dt$$

$P(40) = 10000 + 33057.56459$

There are 133, 058 penguins on the island.

0	1	2	3

The response earns all three of the following points:

1 point: For correct limits

1 point: For the integrand

1 point: For the correct answer

$$P(40) = 100000 + \int_0^{40} (B(t) - D(t))dt$$

$P(40) = 10000 + 33057.56459$

There are 133, 058 penguins on the island.

8-3

20. To the nearest whole number, find the absolute minimum penguin population and the absolute maximum penguin population on the island for $0 \le t \le 40$. Show the analysis that leads to your answer.

Part D

The response can earn up to 4 points:

1 point: For the expression $B(t) - D(t) = 0$

$B(t) - D(t) = 0$

1 point: For the solution for t

$$1000e^{0.06t} = 250e^{0.1t} \Rightarrow t = A = \frac{\ln 4}{0.04} = 34.657359$$

The absolute minimum and absolute maximum occur at a critical point or at an endpoint.

1 point: For the minimum value

1 point: For the maximum value

$P(0) = 100000$

$$P(A) = 100000 + \int_0^A (B(t) - D(t))dt = 139166.667$$

$P(40) = 133058$

The minimum population is 100,000 and the maximum population is 139,167 penguins.

0	1	2	3	4

The response earns all four of the following points:

1 point: For the expression $B(t) - D(t) = 0$

$B(t) - D(t) = 0$

1 point: For the solution for t

$$1000e^{0.06t} = 250e^{0.1t} \Rightarrow t = A = \frac{\ln 4}{0.04} = 34.657359$$

The absolute minimum and absolute maximum occur at a critical point or at an endpoint.

8-3

1 point: For the minimum value

1 point: For the maximum value

$P(0) = 100000$

$$P(A) = 100000 + \int_0^A (B(t) - D(t))dt = 139166.667$$

$P(40) = 133058$

The minimum population is 100,000 and the maximum population is 139,167 penguins.

21. The temperature in a room at midnight is 20 degrees Celsius. Over the next 24 hours, the temperature changes at a rate modeled by the differentiable function H, where $H(t)$ is measured in degrees Celsius per hour and time t is measured in hours since midnight. Which of the following is the best interpretation of $\int_0^6 H(t)\,dt$?

(A) The temperature of the room, in degrees Celsius, at 6:00 A.M.

(B) The average temperature of the room, in degrees Celsius, between midnight and 6:00 A.M.

(C) The change in the temperature of the room, in degrees Celsius, between midnight and 6:00 A.M. ✓

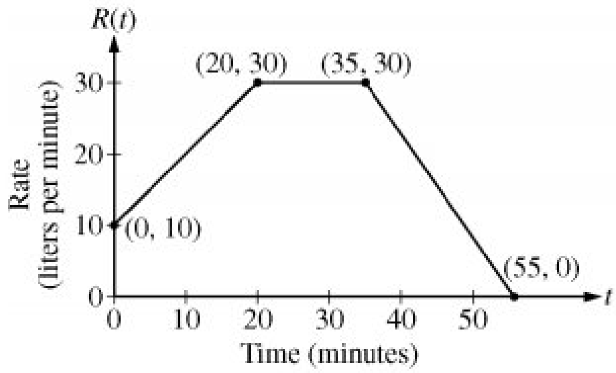
(D) The rate at which the temperature in the room is changing, in degrees Celsius per hour, at 6:00 A.M.

Answer C

This option is correct. The definite integral of the rate of change over an interval gives the net change over that interval. Since $H(t)$ is the rate of change of temperature, the definite integral of $H(t)$ on the interval $0 \le t \le 6$ gives the change in the temperature over that time period.

8-3

22.



At time t = 0 minutes, a tank contains 100 liters of water. The piecewise-linear graph above shows the rate R(t), in liters per minute, at which water is pumped into the tank during a 55-minute period.

At time t = 10 minutes, water begins draining from the tank at a rate modeled by the function D, where $D(t) = 10e^{(\sin t)/10}$ liters per minute. Water continues to drain at this rate until time t = 55 minutes. How many liters of water are in the tank at time t = 55 minutes?

General

1 point is earned for: integral

1 point is earned for: expression for water in the tank

1 point is earned for: answer

$$Amt = 100 + 1150 - \int_0^{55} 10e^{(\sin t)/10} dt$$
$$= 1250 - 450.275371 = 799.725\ (or, 799.724)$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for: integral

1 point is earned for: expression for water in the tank

1 point is earned for: answer

$$Amt = 100 + 1150 - \int_0^{55} 10e^{(\sin t)/10} dt$$
$$= 1250 - 450.275371 = 799.725\ (or, 799.724)$$

8-3

23. A spherical tank contains 81.637 gallons of water at time $t = 0$ minutes. For the next 6 minutes, water flows out of the tank at the rate of $9\sin\left(\sqrt{t+1}\right)$ gallons per minute. How many gallons of water are in the tank at the end of the 6 minutes?

(A) 36.606 ✓

(B) 45.031

(C) 68.858

(D) 77.355

(E) 126.668

24. A function f(t) gives the rate of evaporation of water, in liters per hour, from a pond, where t is measured in hours since 12 noon. Which of the following gives the meaning of $\int_4^{10} f(t)dt$ in the context described?

(A) The total volume of water, in liters, that evaporated from the pond during the first 10 hours after 12 noon

(B) The total volume of water, in liters, that evaporated from the pond between 4 P.M. and 10 P.M. ✓

(C) The net change in the rate of evaporation, in liters per hour, from the pond between 4 P.M. and 10 P.M.

(D) The average rate of evaporation, in liters per hour, from the pond between 4 P.M. and 10 P.M.

(E) The average rate of change in the rate of evaporation, in liters per hour per hour, from the pond between 4 P.M. and 10 P.M.

25. At time $t = 0$ years, a forest preserve has a population of 1500 deer. If the rate of growth of the population is modeled by $R(t) = 2000e^{0.23t}$ deer per year, what is the population at time $t = 3$?

(A) 3987

(B) 5487

(C) 8641

(D) 10,141 ✓

(E) 12,628

26. At time $t = 0$ years, a forest preserve has a population of 1500 deer. If the rate of growth of the population is modeled by $R(t) = 2000e^{0.23t}$ deer per year, what is the population at time $t = 3$?

(A) 3987

(B) 5487

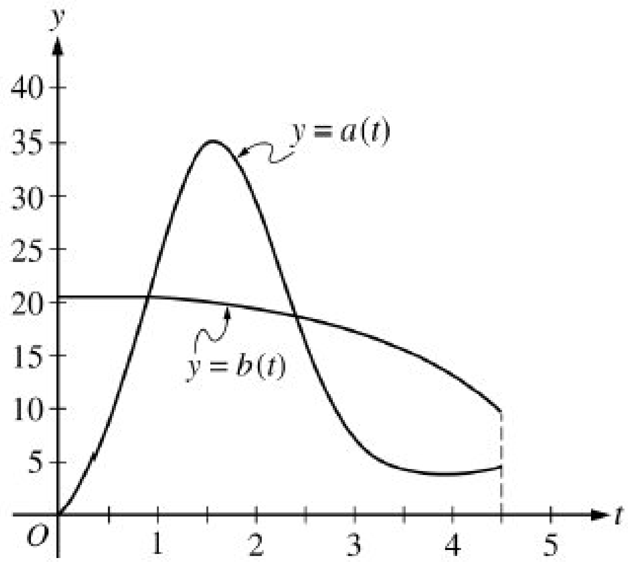
(C) 8641

(D) 10,141 ✓

(E) 12,628

8-3

27. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.



During the time interval $0 \leq t \leq 4.5$ hours, water flows into tank A at a rate of $a\,(t) = (2t - 5) + 5e^{2\sin t}$ liters per hour. During the same time interval, water flows into tank B at a rate of $b\,(t)$ liters per hour. Both tanks are empty at time $t = 0$. The graphs of $y = a(t)$and $y = b(t)$, shown in the figure above, intersect at $t = k$ and $t = 2.416$.

(a) How much water will be in tank A at time $t = 4.5$?

(b) During the time interval $0 \leq t \leq k$ hours, water flows into tank B at a constant rate of 20.5 liters per hour. What is the difference between the amount of water in tank A and the amount of water in tank B at time $t = k$?

(c) The area of the region bounded by the graphs of $y = a(t)$ and $y = b(t)$ for $k \leq t \leq 2.416$ is 14.470. How much water is in tank B at time $t = 2.416$?

(d) During the time interval $2.7 \leq t \leq 4.5$ hours, the rate at which water flows into tank B is modeled by $w\,(t) = 21 - \frac{30t}{(t-8)^2}$ liters per hour. Is the difference $w(t) - a(t)$ increasing or decreasing at time $t = 3.5$? Show the work that leads to your answer.

Part A

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^{4.5} a(t)\;dt = 66.532128$$

At time $t = 4.5$, tank A contains 66.532 liters of water.

0	1	2

8-3

The student response earns all of the following points:

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^{4.5} a(t)\, dt = 66.532128$$

At time $t = 4.5$, tank A contains 66.532 liters of water.

Part B

1 point is earned for: sets $a(k) = 20.5$

1 point is earned for: integral

1 point is earned for: answer

$a(k) = 20.5 \Rightarrow k = 0.892040$

$$\int_0^{k} (20.5 - a(t))\, dt = 10.599191$$

At time $t = k$, the difference in the amounts of water in the tanks is 10.599 liters.

0	1	2	3

The student response earns all of the following points:

1 point is earned for: sets $a(k) = 20.5$

1 point is earned for: integral

1 point is earned for: answer

$a(k) = 20.5 \Rightarrow k = 0.892040$

$$\int_0^{k} (20.5 - a(t))\, dt = 10.599191$$

At time $t = k$, the difference in the amounts of water in the tanks is 10.599 liters.

Part C

1 point is earned for: $\int_k^{2.416} a\,(t)\; dt$

8-3

1 point is earned for: answer

$$\int_0^{2.416} b(t)\,dt = \int_0^k b(t)\,dt + \int_k^{2.416} b(t)\,dt$$

$$\int_0^k b(t)\,dt = 20.5 \cdot k = 18.286826$$

On $k < t < 2.416$, tank A receives $\int_k^{2.416} a\,(t)\,dt = 44.497051$ liters of water, which is 14.470 more liters of water than tank B.

Therefore, $\int_k^{2.416} b\,(t)\,dt = \int_k^{2.416} a\,(t)\,dt - 14.470 = 30.027051$.

$$\int_0^k b\,(t)\,dt + \int_k^{2.416} b\,(t)\,dt = 48.313876$$

At time $t = 2.416$, tank B contains 48.314 (or 48.313) liters of water.

0	1	2

The student response earns all of the following points:

1 point is earned for: $\int_k^{2.416} a\,(t)\,dt$

1 point is earned for: answer

$$\int_0^{2.416} b(t)\,dt = \int_0^k b(t)\,dt + \int_k^{2.416} b(t)\,dt$$

$$\int_0^k b(t)\,dt = 20.5 \cdot k = 18.286826$$

On $k < t < 2.416$, tank A receives $\int_k^{2.416} a\,(t)\,dt = 44.497051$ liters of water, which is 14.470 more liters of water than tank B.

Therefore, $\int_k^{2.416} b\,(t)\,dt = \int_k^{2.416} a\,(t)\,dt - 14.470 = 30.027051$.

$$\int_0^k b\,(t)\,dt + \int_k^{2.416} b\,(t)\,dt = 48.313876$$

At time $t = 2.416$, tank B contains 48.314 (or 48.313) liters of water.

Part D

8-3

1 point is earned for: $w'(3.5) - a'(3.5) < 0$

1 point is earned for: conclusion

$w'(3.5) - a'(3.5) = -1.14298 < 0$

The difference $w(t) - a(t)$ is decreasing at $t = 3.5$.

0	1	2

The student response earns all of the following points:

1 point is earned for: $w'(3.5) - a'(3.5) < 0$

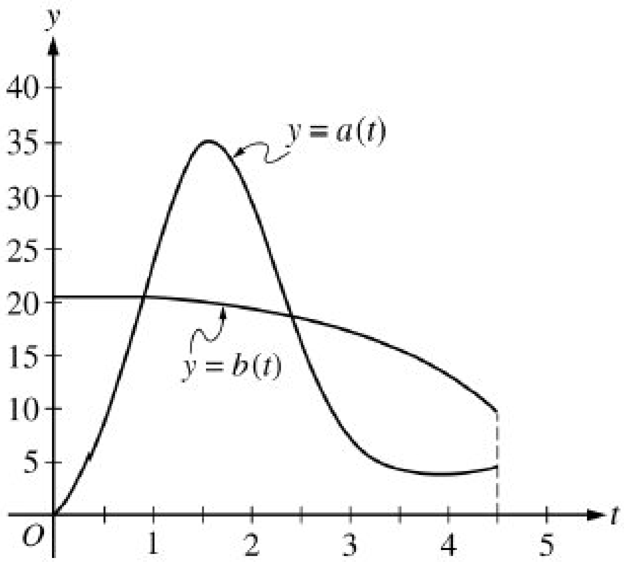
1 point is earned for: conclusion

$w'(3.5) - a'(3.5) = -1.14298 < 0$

The difference $w(t) - a(t)$ is decreasing at $t = 3.5$.

8-3

28. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.



During the time interval $0 \le t \le 4.5$ hours, water flows into tank A at a rate of $a(t) = (2t - 5) + 5e^{2\sin t}$ liters per hour. During the same time interval, water flows into tank B at a rate of $b(t)$ liters per hour. Both tanks are empty at time $t = 0$. The graphs of $y = a(t)$ and $y = b(t)$, shown in the figure above, intersect at $t = k$ and $t = 2.416$.

(a) How much water will be in tank A at time $t = 4.5$?

(b) During the time interval $0 \le t \le k$ hours, water flows into tank B at a constant rate of 20.5 liters per hour. What is the difference between the amount of water in tank A and the amount of water in tank B at time $t = k$?

(c) The area of the region bounded by the graphs of $y = a(t)$ and $y = b(t)$ for $k \le t \le 2.416$ is 14.470. How much water is in tank B at time $t = 2.416$?

(d) During the time interval $2.7 \le t \le 4.5$ hours, the rate at which water flows into tank B is modeled by $w(t) = 21 - \frac{30t}{(t-8)^2}$ liters per hour. Is the difference $w(t) - a(t)$ increasing or decreasing at time $t = 3.5$? Show the work that leads to your answer.

Part A

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^{4.5} a(t)dt = 66.532128$$

At time $t = 4.5$, tank A contains 66.532 liters of water.

0	1	2

8-3

The student response earns all of the following points:

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^{4.5} a(t)dt = 66.532128$$

At time $t = 4.5$, tank A contains 66.532 liters of water.

Part B

1 point is earned for: sets$a(k) = 20.5$

1 point is earned for: integral

1 point is earned for: answer

$$a(k) = 20.5 \Rightarrow k = 0.892040$$
$$\int_0^{k} (20.5 - a(t))dt = 10.599191$$

At time$t = k$, the difference in the amounts of water in the tanks is 10.599 liters.

0	1	2	3

The student response earns all of the following points:

1 point is earned for: sets $a(k) = 20.5$

1 point is earned for: integral

1 point is earned for: answer

$$a(k) = 20.5 \Rightarrow k = 0.892040$$
$$\int_0^{k} (20.5 - a(t))dt = 10.599191$$

At time $t = k$, the difference in the amounts of water in the tanks is 10.599 liters.

Part C

1 point is earned for: $\int_k^{2.416} a(t)dt$

8-3

1 point is earned for: answer

$$\int_0^{2.416} b(t)dt = \int_0^k b(t)dt + \int_k^{2.416} b(t)dt$$

$$\int_0^k b(t)dt = 20.5 \cdot k = 18.286826$$

On $k < t < 2.416$, tank A receives $\int_k^{2.416} a(t)dt = 44.497051$ liters of water, which is 14.470 more liters of water than tank B.

Therefore, $\int_k^{2.416} b(t)dt = \int_k^{2.416} a(t)dt - 14.470 = 30.027051$.

$$\int_0^k b(t)dt + \int_k^{2.416} b(t)dt = 48.313876$$

At time $t = 2.416$, tank B contains 48.314 (or 48.313) liters of water.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for: $\int_k^{2.416} a(t)dt$

1 point is earned for: answer

$$\int_0^{2.416} b(t)dt = \int_0^k b(t)dt + \int_k^{2.416} b(t)dt$$

$$\int_0^k b(t)dt = 20.5 \cdot k = 18.286826$$

On $k < t < 2.416$, tank A receives $\int_k^{2.416} a(t)dt = 44.497051$ liters of water, which is 14.470 more liters of water than tank B.

Therefore, $\int_k^{2.416} b(t)dt = \int_k^{2.416} a(t)dt - 14.470 = 30.027051$.

$$\int_0^k b(t)dt + \int_k^{2.416} b(t)dt = 48.313876$$

At time $t = 2.416$, tank B contains 48.314 (or 48.313) liters of water.

Part D

8-3

1 point is earned for: $w'(3.5) - a'(3.5) < 0$

1 point is earned for: conclusion

$w'(3.5) - a'(3.5) = -1.14298 < 0$

The difference $w(t) - a(t)$ is decreasing at $t = 3.5$.

0	1	2

The student response earns all of the following points:

1 point is earned for: $w'(3.5) - a'(3.5) < 0$

1 point is earned for: conclusion

$w'(3.5) - a'(3.5) = -1.14298 < 0$

The difference $w(t) - a(t)$ is decreasing at $t = 3.5$.

8-3

29.

t (days)	0	10	22	30
$W'(t)$ (GL per day)	0.6	0.7	1.0	0.5

The twice-differentiable function W models the volume of water in a reservoir at time t, where $W(t)$ is measured in (GL) and t is measured in days. The table above gives values of $W'(t)$ sampled at various times during the time interval $0 \le t \le 30$ days. At time $t = 30$, the reservoir contains 125 gigaliters of water.

Use a left Riemann sum, with the three subintervals indicated by the data in the table, to approximate $\int_0^{30} W'(t)\,dt$. Use this approximation to estimate the volume of water $W(t)$, in gigaliters, in the reservoir at time $t = 0$. Show the computations that lead to your answer.

Part B

The response can earn up to 3 points:

1 point: For the left Riemann sum

1 point: For the approximation

1 point: For the correct answer

$$\int_0^{30} W'(t)dt \approx (10)(0.6) + (12)(0.7) + (8)(1.0) = 22.4$$

$$W(0) = W(30) - \int_0^{30} W'(t)dt = 125 - 22.4 = 102.6$$

0	1	2	3 ✓

The response earns all three of the following points:

1 point: For the left Riemann sum

1 point: For the approximation

1 point: For the correct answer

$$\int_0^{30} W'(t)dt \approx (10)(0.6) + (12)(0.7) + (8)(1.0) = 22.4$$

8-3

$$W(0) = W(30) - \int_0^{30} W'(t)dt = 125 - 22.4 = 102.6$$

30. The number of bacteria in a container increases at the rate of $R(t)$ bacteria per hour. If there are 1000 bacteria at time $t = 0$, which of the following expressions gives the number of bacteria in the container at time $t = 3$ hours?

(A) $R(3)$

(B) $1000 + R(3)$

(C) $\int_0^3 R(t)\,dt$

(D) $1000+\int_0^3 R(t)\,dt$ ✓

8-3

31. A tank contains 30 liters of water at time $t = 0$. For time $0 \leq t \leq 20$ minutes, water flows into the tank at a rate modeled by the function *f* defined by $f(t) = 2\ln(t+3) - 1.6$. For time $20 < t \leq 45$ minutes, water flows into the tank at a rate modeled by the function *g* defined by $g(t) = \frac{7.5}{\ln t} + 2.2$. Both $f(t)$ and $g(t)$ are measured in liters per minute.

Part A

Find the rate of change of *g* at time $t = 23$ minutes. Show the setup for your calculations, and indicate units of measure.

Part B

How many liters of water are in the tank at time $t = 20$ minutes? Show the setup for your calculations.

Part C

Let *w* be the function defined by $w(t) = \begin{cases} f(t) & \text{for } 0 \leq t \leq 20 \\ g(t) & \text{for } 20 < t \leq 45 \end{cases}$. Find $\lim\limits_{t \to 20^+} w(t)$. Is *w* continuous at $t = 20$? Use the definition of continuity to justify your answer.

Part D

Consider the function *w* defined in part C. Find the average value of *w* from time $t = 15$ minutes to time $t = 30$ minutes. Show the setup for your calculations.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P1** is earned for a correct value either rounded or truncated to three places after the decimal and for labeling the value as $g'(23)$. An inappropriately rounded answer does not earn the point.
- **P1** can also be earned for the exact value $\dfrac{-7.5}{23(\ln 23)^2}$.

Model Solution:

$g'(23) = -0.033$

0	1

The response accurately includes the criteria below.

- $g'(23)$

8-3

Part A Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P2** is earned for correct units, whether or not they are attached to a numerical value for the rate of change.
- **P2** is also earned for the units “$\text{liters}/\text{minute}^2$.”

Model Solution:

The rate of change in the rate at time $t = 23$ is -0.033 liters per minute per minute.

0	1

The response accurately includes the criteria below.

- Units

Part B Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P3** is earned with a definite integral containing the correct integrand and correct lower and upper limits, with or without the differential.
- If the differential dt is missing:
  - Writing $\int_0^{20} f(t)$ or $30 + \int_0^{20} f(t)$ earns **P3**.
  - Because $\int_0^{20} f(t) + 30$ may mean $\int_0^{20} f(t)dt + 30$ or $\int_0^{20} (f(t) + 30)dt$:
    - Writing $\int_0^{20} f(t) + 30$ in the presence of a correct answer of 95.641 earns **P3**.
    - Writing $\int_0^{20} f(t) + 30$ in the presence of an answer of 95, 96, 95.6, or 95.64 earns **P3**.
    - Writing $\int_0^{20} f(t) + 30$ in the presence of no answer or an incorrect answer does not earn **P3**.
- Writing $\int f(t)dt$, $30 + \int f(t)dt$, $30 + \int f(t)$, or $\int f(t) + 30$ and never providing the definite integral, does not earn **P3**.

Model Solution:

8-3

$$30+\int_0^{20} f(t)dt = 30+\int_0^{20}(2\ln(t+3)-1.6)dt$$

0	1

The response accurately includes the criteria below.

- Definite integral

Part B Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P4** is earned for the correct answer, with or without supporting work. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn the point, unless an earlier point was not earned due to inappropriate rounding.
- Incorrect or unclear communication between the correct integral and the correct answer is treated as scratch work and is not considered in scoring. The following examples earn **P4** for the correct answer:
    - $\int_0^{20} f(t)dt = 65.641060$ so the number of liters of water is 95.641.
    - $\int_0^{20} f(t)dt = 65.641060 + 30 = 95.641060$ (Incorrect linkage is not considered in scoring.)
    - $\int_0^{20} f(t)dt = 95.641060$ (Incorrect linkage is not considered in scoring.)

Model Solution:

There are 95.641 liters of water in the tank at time $t = 20$ minutes.

0	1

The response accurately includes the criteria below.

- Answer

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

8-3

- To earn **P5**, a response must indicate $\lim_{t \to 20^+} w(t) = 4.704$ (or 4.703), either symbolically or verbally. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn **P5**, unless an earlier point was not earned due to inappropriate rounding.
  - A response of "the right-hand limit is 4.704" is sufficient to earn **P5**.

Model Solution:

$$\lim_{t \to 20^+} w(t) = \lim_{t \to 20^+} g(t) = 4.703562$$

The limit as *t* approaches 20 from the right is 4.704 (or 4.703).

0	1

The response accurately includes the criteria below.

- Right-hand limit

Part C Point 6

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P6**, a response must include $w(20) = 4.671$ (or 4.67 or 4.6), $\lim_{t \to 20^+} w(t) \neq w(20)$, and a conclusion of not continuous (with or without indicating $t = 20$).
- Alternate solution:

$$\lim_{t \to 20^-} w(t) = \lim_{t \to 20^-} f(t) = 4.670988$$

Because $\lim_{t \to 20^-} w(t) \neq \lim_{t \to 20^+} w(t)$, *w* is not continuous at $t = 20$.

Such a response is scored analogously to the model solution and earns **P6**.

- Special case: A response of "$\lim_{t \to 20^+} w(t) = 4.7$ and $\lim_{t \to 20^-} w(t) = w(20) = 4.7$, therefore continuous" earns **P6**.

Model Solution:

$$w(20) = f(20) = 4.670988$$

Because $\lim_{t \to 20^+} w(t) \neq w(20)$, *w* is not continuous at $t = 20$.

8-3

0	1

The response accurately includes the criteria below.

- Answer with justification

Part D Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P7**, a response must present one or more definite integrals with correct integrands, with or without differentials. Limits of integration are not considered in scoring for **P7**. That is, any of the following earns **P7**: $\int_a^c w(t)dt$, $\int_a^b w(t)dt + \int_b^c w(t)dt$, or $\int_a^b f(t)dt + \int_b^c g(t)dt$, where a, b, and c are real numbers between 0 and 45, inclusive.

Model Solution:

$$\frac{1}{30-15}\int_{15}^{30} w(t)dt$$

0	1

The response accurately includes the criteria below.

- Definite integral

Part D Point 8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P8**, a response must present a correct definite integral and evidence of division by 15.
  - A response that earned **P7** earns **P8** for $\frac{1}{15}\int_{15}^{30} w(t)dt$, $\frac{1}{15}\left(\int_{15}^{20} w(t)dt + \int_{20}^{30} w(t)dt\right)$, $\frac{1}{15}\left(\int_{15}^{20} f(t)dt + \int_{20}^{30} g(t)dt\right)$, or equivalent.
  - Presenting the correct integral and the correct answer will suffice as evidence of division by 15 and will earn **P8**. Work presented between earning **P7** and the correct answer that does not reveal incorrect work

8-3

done in the calculator will be treated as scratch work and will not be considered in scoring.

- "$\int_{15}^{30} w(t)dt = 67.559$, so 4.504" earns **P8** for correct limits of integration and an implied division by 15 made evident by the correct answer.
- "$\int_{15}^{30} w(t)dt = \frac{67.559}{15} = 4.504$" earns **P8** for correct limits of integration and an implied division by 15 made evident by the correct answer. The incorrect communication between the integral and the answer does not reveal incorrect work done in the calculator. This work is treated as scratch work and is not considered in scoring of **P8**.

- A response that did not earn **P7** is eligible to earn **P8** for $\frac{1}{15}\int_{15}^{30} f(t)dt$ or $\frac{1}{15}\int_{15}^{30} g(t)dt$.

Model Solution:

$$\frac{1}{15}\left(\int_{15}^{20} w(t)dt + \int_{20}^{30} w(t)dt\right)$$
$$= \frac{1}{15}\left(\int_{15}^{20} f(t)dt + \int_{20}^{30} g(t)dt\right)$$

0	1

The response accurately includes the criteria below.

- Average value formula

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P9** is earned for the correct answer, with or without supporting work. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn the point, unless an earlier point was not earned due to inappropriate rounding.

Model Solution:

The average value of w from time $t = 15$ and time $t = 30$ minutes is 4.504 (or 4.503) liters per minute.

0	1

The response accurately includes the criteria below.

8-3

- Answer

8-3

32. A store is having a 12-hour sale. The total number of shoppers who have entered the store t hours after the sale begins is modeled by the function S defined by $S(t) = 0.5t^4 - 16t^3 + 144t^2$ for $0 \le t \le 12$. At time $t = 0$, when the sale begins, there are no shoppers in the store.

(a) At what rate are shoppers entering the store 3 hours after the start of the sale?

(b) Find the value of $\frac{1}{3}\int_6^9 S'(t)\,dt$. Using correct units, explain the meaning of $\frac{1}{3}\int_6^9 S'(t)\,dt$ in the context of this problem.

(c) The rate at which shoppers leave the store, measured in shoppers per hour, is modeled by the function L defined by $L(t) = -80 + \frac{4400}{t^2 - 14t + 55}$ for $0 \le t \le 12$. According to the model, how many shoppers are in the store at the end of the sale (time $t = 12$)? Give your answer to the nearest whole number.

(d) Using the given models, find the time t, $0 \le t \le 12$, at which the number of shoppers in the store is the greatest. Justify your answer.

Part A

The response can earn up to 1 point:

1 point is earned for the answer

$S'(3) = 486$ shoppers/hour

0	1

The student earns all of the following point:

1 point is earned for the answer

$S'(3) = 486$ shoppers/hour

Part B

The response can earn up to 2 points:

1 point is earned for the value of $\frac{1}{3}\int_6^9 S'(t)\,dt$

$$\frac{1}{3}\int_6^9 S'(t)\,dt = \frac{1}{3}(S(9) - S(6))$$

$$= \frac{1}{3}(3280.5 - 2376) = 301.5$$

1 point is earned for the meaning of $\frac{1}{3}\int_6^9 S'(t)\,dt$

8-3

$\frac{1}{3}\int_6^9 S'(t)\,dt$ is the average rate at which shoppers are entering the store in shoppers/hr between times $t = 6$ and $t = 9$ hours.

0	1	2

The student earns all of the following points:

1 point is earned for the value of $\frac{1}{3}\int_6^9 S'(t)\,dt$

$$\begin{aligned}\frac{1}{3}\int_6^9 S'(t)\,dt &= \frac{1}{3}(S(9) - S(6)) \\ &= \frac{1}{3}(3280.5 - 2376) = 301.5\end{aligned}$$

1 point is earned for the meaning of $\frac{1}{3}\int_6^9 S'(t)\,dt$

$\frac{1}{3}\int_6^9 S'(t)\,dt$ is the average rate at which shoppers are entering the store in shoppers/hr between times $t = 6$ and $t = 9$ hours.

Part C

The response can earn up to 3 points:

1 point is earned for the integral

1 point is for use of $S(12)$

1 point is for the answer

$$S(12) - \int_0^{12} L(t)\,dt = 195.701684$$

Therefore, there are approximately 196 shoppers in the store at the end of the sale.

0	1	2	3

The student earns all of the following points:

1 point is earned for the integral

8-3

1 point is for use of $S(12)$

1 point is for the answer

$$S(12) - \int_0^{12} L(t)\, dt = 195.701684$$

Therefore, there are approximately 196 shoppers in the store at the end of the sale.

Part D

The response can earn up to 3 points:

1 point is earned for considering $S'(t) - L(t) = 0$

1 point is earned for the answer

1 point is earned for justification

The number of shoppers in the store at time t is given by the function N defined by

$$N(t) = S(t) - \int_0^{t} L(x)\, dx$$

$N'(t) = S'(t) - L(t) = 0$ when $t = 5.545066$

t	$N(t)$
0	0
5.545066	1361.832842
12	195.702

The number of shoppers in the store is greatest at time $t = 5.545$ hours.

0	1	2	3

The student earns all of the following points:

1 point is earned for considering $S'(t) - L(t) = 0$

1 point is earned for the answer

1 point is earned for justification

The number of shoppers in the store at time t is given by the function N defined by

8-3

$$N(t) = S(t) - \int_0^t L(x)\,dx$$

$N'(t) = S'(t) - L(t) = 0$ when $t = 5.545066$

t	$N(t)$
0	0
5.545066	1361.832842
12	195.702

The number of shoppers in the store is greatest at time $t = 5.545$ hours.

33. For time $t \geq 0$, the acceleration of an object moving in a straight line is given by $a(t) = \sin\left(\frac{t^2}{3}\right)$. What is the net change in velocity from time $t = 0.75$ to time $t = 2.25$?

(A) -0.665

(B) 0.656

(C) 0.807

(D) 0.984 ✓

Answer D

Correct. The definite integral of the rate of change over an interval gives the net change over the interval. Therefore, the net change in velocity over the interval $[0.75, 2.25]$ is $\int_{0.75}^{2.25} v'(t)\,dt = \int_{0.75}^{2.25} a(t)\,dt = \int_{0.75}^{2.25} \sin\left(\frac{t^2}{3}\right) dt$. The calculator is used to find the numerical value of this definite integral.

8-3

34. For time $t \geq 0$, the acceleration of an object moving in a straight line is given by $a(t) = \ln\left(3 + t^4\right)$. What is the net change in velocity from time $t = 1$ to time $t = 5$?

(A) -0.204

(B) 4.217

(C) 5.056

(D) 16.868 ✓

Answer D

Correct. The definite integral of the rate of change over an interval gives the net change over the interval. Therefore, the net change in velocity over the interval $[1, 5]$ is $\int_1^5 v'(t)\,dt = \int_1^5 a(t)\,dt = \int_1^5 \ln\left(3 + t^4\right) dt$. The calculator is used to find the numerical value of this definite integral.

35. The furnace in a home consumes heating oil during a particular month at a rate modeled by the function F given by $F(t) = 0.3 + 0.1t - 0.85\cos\left(\frac{2\pi}{15}(t + 5)\right)$, where $F(t)$ is measured in gallons per day and t is the number of days since the start of the month. How many gallons of oil does the furnace consume during the first 14 days of the month (from $t = 0$ to $t = 14$) ?

(A) 10.150

(B) 13.739 ✓

(C) 17.597

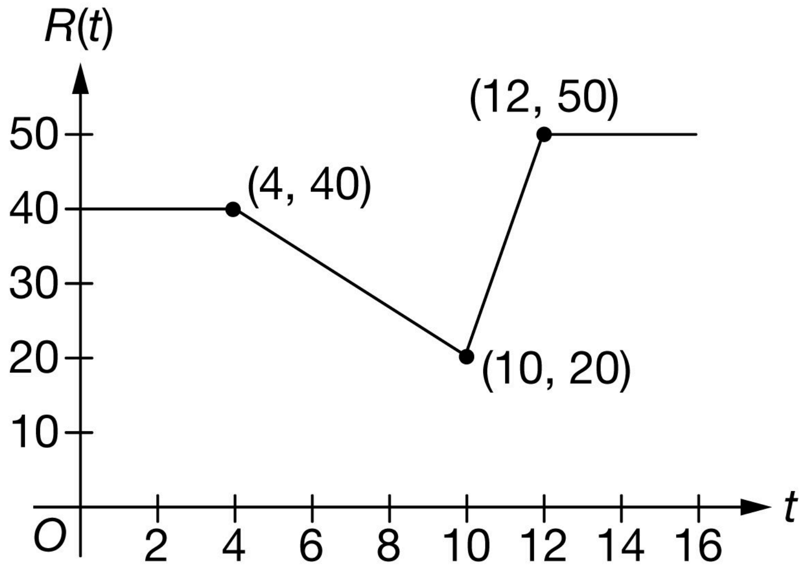
(D) 25.044

Answer B

Correct. The definite integral of the rate of change over an interval gives the net change over that interval. Since $F(t)$ is the rate at which oil is consumed, the definite integral of $F(t)$ on the interval $0 \leq t \leq 14$ gives the total amount of oil that is consumed during the first 14 days. The graphing calculator is used to numerically evaluate the definite integral. $\int_0^{14} \left(0.3 + 0.1t - 0.85\cos\left(\frac{2\pi}{15}(t + 5)\right)\right) dt = 13.739$

8-3

36.



During a five-kilometer road race, runners pass a viewing stand. The rate at which runners pass the viewing stand over a certain 16-minute time interval is modeled by the continuous function R, where $R(t)$ is measured in runners per minute and t is measured in minutes. The graph of R is shown in the figure above. How many runners pass the viewing stand between time $t = 0$ and time $t = 16$?

(A) 50

(B) 560

(C) 610

(D) 640

Answer C

Correct. The definite integral of the rate of change over an interval gives the net change over that interval. Since $R(t)$ is the rate at which runners pass the viewing stand, the definite integral of $R(t)$ over the interval $0 \leq t \leq 16$ gives the total number of runners that pass over that time period. The value of the definite integral can be found geometrically by computing the area between the graph of $R(t)$ and the t-axis.

$$\begin{aligned}\int_0^{16} R(t)\ dt &= \int_0^4 R(t)\ dt + \int_4^{10} R(t)\ dt + \int_{10}^{12} R(t)\ dt + \int_{12}^{16} R(t)\ dt \\ &= 4(40) + 6\left(\tfrac{40+20}{2}\right) + 2\left(\tfrac{20+50}{2}\right) + 4(50) \\ &= 160 + 180 + 70 + 200 = 610\end{aligned}$$

8-3

37. The rate at which sand is poured into a bag is modeled by the function r given by $r(t) = 15t - 2t^2$, where $r(t)$ is measured in milliliters per second and t is measured in seconds after the sand begins pouring. How many milliliters of sand accumulate in the bag from time $t = 0$ to time $t = 2$?

(A) 7

(B) 22

(C) $\frac{74}{3}$

(D) 44

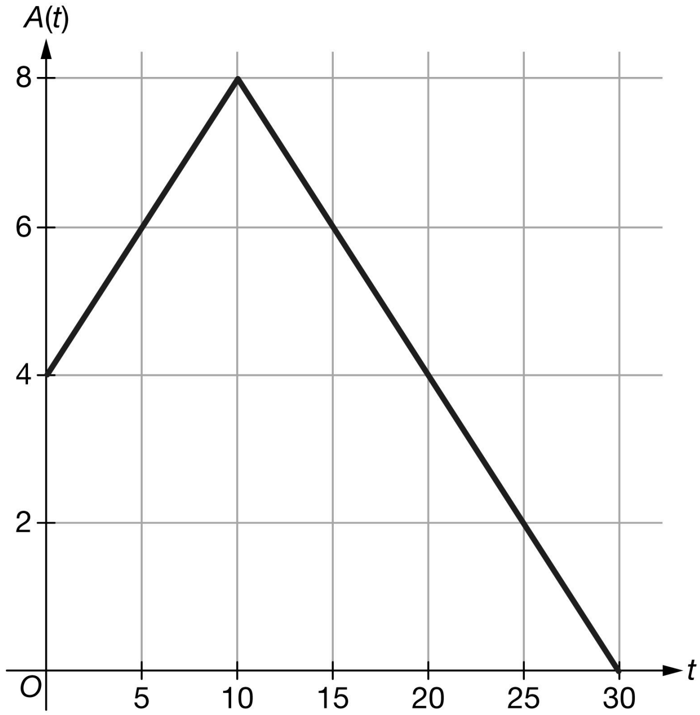
Answer C

Correct. The definite integral of the rate of change over an interval gives the net change over that interval. Since r is the rate at which sand is accumulating, the definite integral of r on the interval $0 \le t \le 2$ gives the total accumulation of sand over that time period, as follows.

$$\int_0^2 r(t)\,dt = \int_0^2 15t - 2t^2\,dt = 15\tfrac{t^2}{2} - \tfrac{2t^3}{3}\Big|_0^2 = 15\left(\tfrac{4}{2}\right) - \tfrac{2}{3}(8) - [0 - 0] = \tfrac{74}{3}$$

8-3

38.



The rate at which ants arrive at a picnic is modeled by the function A, where $A(t)$ is measured in ants per minute and t is measured in minutes. The graph of A for $0 \leq t \leq 30$ is shown in the figure above. How many ants arrive at the picnic during the time interval $0 \leq t \leq 30$?

(A) 8

(B) 70

(C) 120

(D) 140

Answer D

Correct. The definite integral of the rate of change over an interval gives the net change over that interval. Since $A(t)$ is the rate at which ants are arriving, the definite integral of $A(t)$ on the interval $0 \leq t \leq 30$ gives the total number of ants that arrive at the picnic over that time period. The value of the definite integral can be found geometrically by computing the areas of the trapezoid and triangle between the graph of $A(t)$ and the t-axis, as follows.

$$\int_0^{30} A(t)\,dt = 10 \cdot \left(\tfrac{4+8}{2}\right) + \tfrac{1}{2} \cdot 20 \cdot 8 = 60 + 80 = 140$$

One can also count the boxes in the grid, taking into account the half boxes, to get a total of 14 boxes, each of area 10, for a total area of 140.

8-3

39. The rate at which rain accumulates in a bucket is modeled by the function r given by $r(t) = 10t - t^2$, where $r(t)$ is measured in milliliters per minute and t is measured in minutes since the rain began falling. How many milliliters of rain accumulate in the bucket from time $t = 0$ to time $t = 3$?

(A) 4

(B) 21

(C) 36

(D) 63

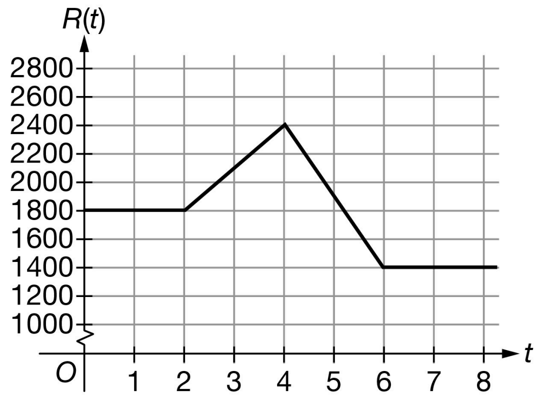
Answer C

Correct. The definite integral of the rate of change over an interval gives the net change over that interval. Since $r(t)$ is the rate at which rain is accumulating, the definite integral of $r(t)$ on the interval $0 \le t \le 3$ gives the total accumulation of rain over that time period.

$$\int_0^3 r(t)\,dt = \int_0^3 \left(10t - t^2\right) dt = \left(5t^2 - \tfrac{1}{3}t^3\right)\Big|_0^3 = (45 - 9) - (0 - 0) = 36$$

8-3

40.



A traffic engineer developed the continuous function R, graphed above, to model the rate at which vehicles pass a certain intersection over an 8-hour time period, where $R(t)$ is measured in vehicles per hour and t is the number of hours after 6:00 AM. According to the model, how many vehicles pass the intersection between time $t = 0$ and time $t = 8$?

(A) 1400

(B) 1600

(C) 14,400

(D) 44,800

Answer C

Correct. The definite integral of the rate of change over an interval gives the net change over that interval. Since R is the rate at which vehicles pass the intersection, the definite integral of R over the interval $0 \le t \le 8$ gives the total number of vehicles that pass over that time period. The value of the definite integral can be found geometrically by computing the area between the graph of R and the t-axis.

$$\int_0^8 R(t)\,dt = \int_0^2 R(t)\,dt + \int_2^4 R(t)\,dt + \int_4^6 R(t)\,dt + \int_6^8 R(t)\,dt$$

$$= (2)(1800) + \tfrac{1}{2}(1800 + 2400)(2) + \tfrac{1}{2}(2400 + 1400)(2) + 2(1400)$$

$$= 14{,}400$$

8-3

41. Water is leaking from a dam during a particular week at a rate modeled by the function F given by $F(t) = 5 + \frac{t}{4} - \sin\left(\frac{t^2}{5}\right)$, where $F(t)$ is measured in gallons per day and t is the number of days since the start of the week on Sunday. How many gallons of water leak from the dam Tuesday through Thursday, days 2 through 4 ?

(A) 9.565

(B) 10.010 ✓

(C) 10.841

(D) 12.117

Answer B

Correct. The definite integral of the rate of change over an interval gives the net change over that interval. Since F gives the rate, in gallons, at which water leaks from the dam, the definite integral of $F(t)$ on the interval $2 \leq t \leq 4$ gives the number of gallons of water that leak from the dam on those days. The graphing calculator is used to numerically evaluate the definite integral.

$$\int_2^4 \left(5 + \frac{t}{4} - \sin\left(\frac{t^2}{5}\right)\right) dt = 10.010$$

42. A cup of coffee is poured, and the temperature is measured to be 120 degrees Fahrenheit. The temperature of the coffee then decreases at a rate modeled by $r(t) = 55e^{-0.03t^2}$ degrees Fahrenheit per minute, where t is the number of minutes since the coffee was poured. What is the temperature of the coffee, in degrees Fahrenheit, at time $t = 1$ minute?

(A) 53.4°F

(B) 54.5°F

(C) 65.5°F ✓

(D) 66.6°F

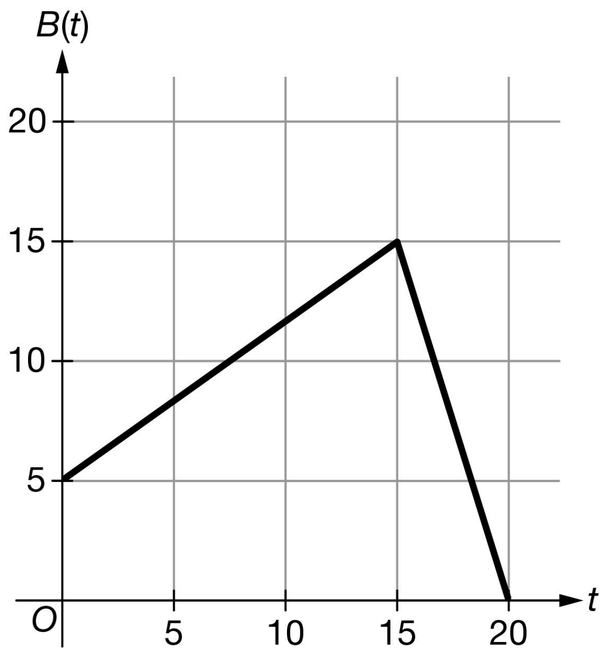
Answer C

Correct. The definite integral of the rate of change over an interval gives the net change over that interval. Since $r(t)$ is the rate at which the temperature is decreasing, the definite integral of $r(t)$ on the interval $0 \leq t \leq 1$ gives the total decrease in the coffee's temperature over that time period. This net change must be subtracted from the coffee's temperature at $t = 0$ to get the temperature at $t = 1$, as follows. The calculator is used to do the numerical integration.

$$120 - \int_0^1 r(t)\, dt = 120 - \int_0^1 55e^{-0.03t^2} dt = 65.5$$

8-3

43.



The rate at which people arrive at a theater box office is modeled by the function B, where $B(t)$ is measured in people per minute and t is measured in minutes. The graph of B for $0 \le t \le 20$ is shown in the figure above. Which of the following is closest to the number of people that arrive at the box office during the time interval $0 \le t \le 20$?

(A) 15

(B) 38

(C) 150

(D) 188

Answer D

Correct. The definite integral of the rate of change over an interval gives the net change over that interval. Since $B(t)$ is the rate at which people are arriving, the definite integral of $B(t)$ on the interval $0 \le t \le 20$ gives the total number of people that arrive at the box office over that time period. The value of the definite integral can be found geometrically by computing the areas of the trapezoid and triangle between the graph of B and the t-axis, as follows.

$\int_0^{20} B(t)\,dt = \frac{1}{2}(15+5)(15) + \frac{1}{2}(15)(5) = \frac{375}{2}$. This rounds to 188.

8-3

44. A ceramic mug is removed from a pottery kiln once the mug reaches a temperature of 700 degrees Celsius. The temperature of the mug then decreases at a rate given by $r\,(t) = \frac{20+100\arctan t}{1+t}$ degrees Celsius per minute, where t is the number of minutes since being removed from the kiln. What is the temperature of the mug, to the nearest degree Celsius, at time $t = 10$ minutes?

(A) 15

(B) 288

(C) 412 ✓

(D) 685

Answer C

Correct. The definite integral of the rate of change over an interval gives the net change over that interval. Since $r(t)$ is the rate at which the temperature is decreasing, the definite integral of $r(t)$ on the interval $0 \le t \le 10$ gives the total decrease in the mug's temperature over that time period. This net change must be subtracted from the mug's temperature at $t = 0$ to get the temperature at $t = 10$ as follows. The calculator is used to do the numerical integration.

$$700 - \int_0^{10} r(t)\,dt = 700 - \int_0^{10} \frac{20 + 100\arctan t}{1+t}\,dt = 411.897$$

8-3

45. The temperature inside a house t hours after midnight $(t = 0)$ is given by the twice-differentiable function H, where $H(t)$ is measured in degrees Fahrenheit. Which of the following is an interpretation of the expression $H(0) + \int_0^8 H'(t)\,dt$?

(A) The temperature, in degrees Fahrenheit, inside the house at 8 A.M. $(t = 8)$

(B) The average temperature, in degrees Fahrenheit, inside the house between midnight and 8 A.M. $(t = 8)$

(C) The change in temperature, in degrees Fahrenheit, inside the house between midnight and 8 A.M. $(t = 8)$

(D) The average rate of change of the temperature, in degrees Fahrenheit per hour, inside the house between midnight and 8 A.M. $(t = 8)$

Answer A

Correct. The definite integral of the rate of change of the temperature is the total change of the temperature between midnight $(t = 0)$ and 8 A.M. $(t = 8)$. Since $H(0)$ is the temperature at midnight, adding the total change to $H(0)$ will give the temperature at 8 A.M.

Alternatively, by the Fundamental Theorem of Calculus,

$H(0) + \int_0^8 H'(t)dt = H(0) + (H(8) - H(0)) = H(8)$, which is the temperature inside the house at 8 A.M.

8-3

46. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	2	5	9	11	12
$L(t)$ (cars per hour)	15	40	24	68	18

The rate at which cars enter a parking lot is modeled by $E(t) = 30 + 5(t-2)(t-5)e^{-0.2t}$. The rate at which cars leave the parking lot is modeled by the differentiable function L. Selected values of $L(t)$ are given in the table above. Both $E(t)$ and $L(t)$ are measured in cars per hour, and time t is measured in hours after 5 A.M. ($t = 0$). Both functions are defined for $0 \le t \le 12$.

(a) What is the rate of change of $E(t)$ at time $t = 7$? Indicate units of measure.

(b) How many cars enter the parking lot from time $t = 0$ to time $t = 12$? Give your answer to the nearest whole number.

(c) Use a trapezoidal sum with the four subintervals indicated by the data in the table to approximate $\int_2^{12} L(t)\,dt$. Using correct units, explain the meaning of $\int_2^{12} L(t)\,dt$ in the context of this problem.

(d) For $0 \le t < 6$, 5 dollars are collected from each car entering the parking lot. For $6 \le t \le 12$, 8 dollars are collected from each car entering the parking lot. How many dollars are collected from the cars entering the parking lot from time $t = 0$ to time $t = 12$? Give your answer to the nearest whole dollar.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1

8-3

The student response accurately includes the criteria below.

☐ answer with units

Solution:

$E'(7) = 6.164924$

The rate of change of $E(t)$ at time $t = 7$ is 6.165 (or 6.164) cars per hour per hour.

Part B

The first point is earned with $E(t)$ or $30 + 5(t - 2)(t - 5)e^{-0.2t}$ written as the integrand. Limits are part of the second point.

It is possible to earn the second point with a single copy error of a constant in the integrand.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

☐ integrand

☐ answer

Solution:

$$\int_0^{12} E(t)\,dt = 520.070489$$

To the nearest whole number, 520 cars enter the parking lot from time $t = 0$ to time $t = 12$.

Part C

The first point is earned for showing the method of a trapezoidal sum, so the sum of three products must be given. This point can be earned with a maximum of one error in either a coefficient or a value extracted from the table.

The second point can be earned concurrently with the first point. Simplification is not required. Functions values from the table must be substituted.

The third point requires that the explanation includes (1) units (cars), (2) the time interval ($2 \le t \le 12$ or 7 A.M. to 5 P.M), (3) a meaning that expresses the (approximate) number of cars leaving the parking lot.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

8-3

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ trapezoidal sum
- ☐ approximation
- ☐ explanation

Solution:

$$\int_2^{12} L(t)\,dt \approx (5-2)\cdot\frac{L(2)+L(5)}{2}+(9-5)\cdot\frac{L(5)+L(9)}{2}$$
$$+(11-9)\cdot\frac{L(9)+L(11)}{2}+(12-11)\cdot\frac{L(11)+L(12)}{2}$$
$$=3\cdot\frac{15+40}{2}+4\cdot\frac{40+24}{2}+2\cdot\frac{24+68}{2}+1\cdot\frac{68+18}{2}$$
$$=345.5$$

$\int_2^{12} L(t)\,dt$ is the number of cars that leave the parking lot in the 10 hours between 7 A.M. $(t=2)$ and 5 P.M. $(t=12)$.

Part D

The first point is earned with two definite integrals, both having $E(t)$ as the integrand.

The third point may be earned with an answer correctly rounded or truncated to 1, 2, or 3 decimal places.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ integrand
- ☐ limits and constants
- ☐ answer

Solution:

8-3

$$5\int_0^6 E(t)\,dt + 8\int_6^{12} E(t)\,dt = 3530.1396$$

To the nearest dollar, 3530 dollars are collected from time $t = 0$ to time $t = 12$.

8-3

47. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (years)	0	4	9	12
$R(t)$ (bears per year)	150	156	159	161

The rate at which the number of bears in a region is changing is modeled by the differentiable function R, where $R(t)$ is measured in bears per year and t is measured in years. Selected values of $R(t)$ are shown in the table.

(a) Approximate $R'(7)$ using the average rate of change of R over the interval $4 \le t \le 9$. Show the computations that lead to your answer, and indicate units of measure.

(b) Interpret the meaning of $\frac{1}{12}\int_0^{12} R(t)\,dt$ in the context of the problem. Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\frac{1}{12}\int_0^{12} R(t)\,dt$.

(c) The rate at which the number of bears in the region is changing can also be modeled by the function P defined by $P(t) = 150e^{0.02\sqrt{t}}$, where $P(t)$ is measured in bears per year and t is measured in years. There are 300 bears in the region at time $t = 0$. Use this model to find the number of bears in the region at time $t = 12$. Show the setup for your calculations, and give your answer to the nearest whole number of bears.

(d) Using the model defined in part (c), find the time t, for $0 \le t \le 12$, at which the number of bears in the region is a maximum. Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

8-3

To earn the first point a response must provide a numerical difference and a quotient. A response of only or earns the first point.

To earn the second point, a response must have "bear per year per year" (or the equivalent) attached to a numerical value.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Approximation with supporting work
- [ ] Units

Solution:

$$R'(7) \approx \frac{R(9)-R(4)}{9-4} = \frac{159-156}{5} = 0.6$$

$R'(7) \approx 0.6$ bear per year per year

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, a response must provide "average," "rate of change," and "from time $t = 0$ to $t = 12$" (or equivalent). Units are not required to earn the first point.

To earn the second point, at least five of the six factors in the Riemann sum must be correct. The factor of $\frac{1}{12}$ is assessed in the third point.

If there is any error in the Riemann sum, the response does not earn the third point.

A response of only $\frac{1}{12}(150 \cdot 4 + 156 \cdot 5 + 159 \cdot 3)$ earns the second and third points.

A response that presents the correct average value with accompanying work that does not explicitly present the six factors and/or the sum process does not earn the second point but will earn the third point. For example, responses of "$\frac{150 \cdot 4+156 \cdot 5+477}{12}$" or "$600, 780, 477 \rightarrow \frac{1857}{12}$" would earn the third point but not the second.

A response that presents an answer of $\frac{156 \cdot 4+159 \cdot 5+161 \cdot 3}{12}$ (which uses a right Riemann sum) earns 1 of the last 2 points. If this response is simplified, the result must be $\frac{1902}{12}$, $\frac{951}{6}$, $\frac{317}{2}$, or 158.5.

An unsupported response of $\frac{1857}{12}$, $\frac{619}{4}$, or 154.75 does not earn the second or third point.

8-3

A response that miscommunicates by equating two unequal quantities, e.g., $\int_0^{12} R(t)\ dt = \frac{150 \cdot 4 + 156 \cdot 5 + 159 \cdot 3}{12}$, does not earn the third point.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Interpretation
- ☐ Form of left Riemann sum
- ☐ Answer

Solution:

$\frac{1}{12}\int_0^{12} R(t)\ dt$ is the average rate of change in the number of bears in the region, in bears per year, from time $t = 0$ to time $t = 12$ years.

$$\int_0^{12} R(t)\ dt \approx R(0)\cdot(4-0) + R(4)\cdot(9-4) + R(9)\cdot(12-9)$$
$$= 150\cdot(4-0) + 159\cdot(9-4) + 159\cdot(12-9)$$
$$= 150\cdot 4 + 156\cdot 5 + 159\cdot 3 = 1857$$

$$\frac{1}{12}\int_0^{12} R(t)\ dt \approx \frac{1857}{12} = \frac{619}{4} = 154.75 \text{ bears per year}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for a definite integral $\int_a^b P(t)\ dt$ with any limits of integration. If the limits are incorrect, the response does not earn the second point.

The second point cannot be earned without the first point, and is earned only for an answer of 2185 or 2185.339 or 2185.338.

Missing differentials:

- A response of only $\int_0^{12} P(t)$ or $\int_0^{12} P(t) = 1885.3389$ earns the first point but not the second.

8-3

- A response of $300 + \int_0^{12} P(t)$ earns the first point and in the presence of the correct answer earns the second point.
- A response of $\int_0^{12} P(t) \; + 300$ does not earn the first point but earns the second point for the correct answer.

A response that miscommunicates, e.g., $\int_0^{12} P(t)\, dt = 1885.3389 + 300$, does not earn the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Integral
- ☐ Answer

Solution:

$$300 + \int_0^{12} P(t)\, dt = 300 + 1885.3389 = 2185.3389$$

At $t = 12$ years, there are 2185 bears in the region.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that considers $P = 0$ earns the first point.

A response of “at time $t = 12$ ” with no justification does not earn the second point.

The second point cannot be earned for any answer other than time $t = 12$.

Alternate solution using Candidates Test:

$P(t) = 0$ has no solutions in the interval $0 \le t \le 12$, so the maximum number of bears occurs at an endpoint of the interval.

The region has 300 bears at time $t = 0$.

From part (c), the region has 2185 bears at time $t = 12$.

For $0 \le t \le 12$, the maximum number of bears is at time $t = 12$ years.

8-3

- The scoring is the same as the model solution.
- A response using this argument may import an incorrect value for the number of bears in the region in year 12 from part (c), provided that value is larger than 300.
- A response using this argument need not report the number of bears at time $t = 12$. Stating that this number is greater than 300 is sufficient.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Considers sign of P
- [ ] Answer with justification

Solution:

$P(t) > 0$ for all $t \geq 0$, so the bear population is always increasing.

For $0 \leq t \leq 12$, the maximum number of bears is at time $t = 12$ years.

8-3

48. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (years)	0	10	22	30	40
$A(t)$ (hundred pounds per year)	163	142	124	118	113

The amount of nitrogen in a particular river is affected by two sources. Nitrogen from the first source enters the river at a rate modeled by the decreasing, differentiable function A. Selected values of $A(t)$ are shown in the table above. Nitrogen from the second source enters the river at a rate modeled by the function B defined by $B(t) = 20 + \left(\frac{95}{1+\sqrt[3]{t}}\right)^{1.25}$. Both $A(t)$ and $B(t)$ are measured in hundreds of pounds of nitrogen per year, and t is measured in years for $0 \leq t \leq 40$.

(a) Using a right Riemann sum with the four subintervals indicated by the data in the table, approximate the total amount of nitrogen from the first source that enters the river over the time interval $0 \leq t \leq 40$. Is this approximation an overestimate or an underestimate? Give a reason for your answer.

(b) Find $B'(39)$. Using correct units, interpret the meaning of $B'(39)$ in the context of the problem.

(c) Based on model B, find the average rate, in hundreds of pounds per year, at which nitrogen from the second source enters the river during the time interval $5 \leq t \leq 35$.

(d) Using model B and your approximation from part (a), find an approximation for the total combined amount of nitrogen from the two sources that enters the river over the time interval $0 \leq t \leq 40$. Give your answer to the nearest hundred pounds.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

8-3

The first point is earned for a sum of four products with at most one error among the 8 factors. If there is an error in the setup, the second point is not earned.

A response of $10 \cdot A(10) + 12 \cdot A(22) + 8 \cdot A(30) + 10 \cdot A(40)$ would earn the first point, but not yet the second. The first two points are earned for $10 \cdot 142 + 12 \cdot 124 + 8 \cdot 118 + 10 \cdot 113$.

Without the products, the sum $1420 + 1488 + 944 + 1130$ by itself would earn the second point but not the first point.

Once the first point is earned it cannot be lost by incorrectly using values from the table or performing some type of operation on the Riemann sum.

A completely correct setup and approximation using a left Riemann sum $(10 \cdot 163 + 12 \cdot 142 + 8 \cdot 124 + 10 \cdot 118 = 5506)$ will earn one of the first two points and is not eligible for the third point. Any error in the setup or evaluation using a left Riemann sum will earn no points.

Units are not scored in this part, except an approximation of $498{,}200$ must clearly be labeled as *pounds* to earn the second point.

We will interpret the word "function" to mean A, unless otherwise prescribed.

The response must indicate which function is decreasing. A response of "underestimate because it (or the graph, or the curve) is decreasing" will not earn the third point.

The third point will not be earned if the response concludes the approximation is an underestimate because the table values are decreasing.

The third point will not be earned if incorrect statements are made in addition to saying A is decreasing.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Right Riemann sum setup
- ☐ Approximation
- ☐ Underestimate with reason

Solution:

$$\int_0^{40} A(t)\,dt \approx (10-0) \cdot A(10) + (22-10) \cdot A(22)$$
$$+\ (30-22) \cdot A(30) + (40-30) \cdot A(40)$$

$$= 10 \cdot 142 + 12 \cdot 124 + 8 \cdot 118 + 10 \cdot 113 = 4982$$

8-3

Approximately 4982 hundred pounds of nitrogen from the first source enters the river over the time interval $0 \leq t \leq 40$.

The approximation is an underestimate because a right Riemann sum is used and A is decreasing.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The value of $B'(39)$ may appear in the interpretation. No work is required in order to earn the first point.

In the presence of a correct value for $B'(39)$, any symbolic attempt at differentiation will be ignored.

Any unevaluated expression equivalent to $1.25\left(\frac{95}{1+\sqrt[3]{39}}\right)^{0.25}\left(\frac{-95}{\left(1+\sqrt[3]{39}\right)^2}\right)\left(\frac{1}{3}(39)^{-2/3}\right)$ will earn the first point.

The second point cannot be earned if no value for $B'(39)$ is presented.

The units (hundred pounds per year per year) may be attached to the value of $B'(39)$ and if so, they do not need to be repeated in the interpretation.

If units attached to $B'(39)$ do not agree with units in the interpretation, read the units in the interpretation.

The interpretation must include time $t = 39$, rate of change of the rate at which nitrogen enters the river (or that the rate that nitrogen enters the river is decreasing at the stated rate), and units of hundred pounds per year per year.

The second point cannot be earned with an incorrect statement such as "… is decreasing at a rate of -0.385 …" or the statement "… is increasing at a rate of -0.385 …."

Use of the term "acceleration" of the amount of nitrogen coming from the second source will not earn the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Value
- ☐ Interpretation

Solution:

$B'(39) = -0.384963$

8-3

At time $t = 39$ years, the rate at which nitrogen from the second source enters the river was decreasing at a rate of 0.385 (or 0.384) hundred pounds per year per year.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for presenting the definite integral $\int_5^{35} B(t)\, dt$.

Dividing by 30 will be scored in the second point.

Once the first point is earned, it cannot be lost.

If the first point is not earned because of a copy error in the integrand, the response can earn the second point for the correct answer.

An unsupported answer of 80.063 does not earn the second point.

Units are not scored in this part, except an answer of 8006.3 must clearly be labeled as *pounds per year* in order to earn the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Definite integral
- ☐ Answer

Solution:

$$\frac{1}{35-5}\int_5^{35} B(t)\, dt = 80.063385$$

During the time interval $5 \leq t \leq 35$, the average rate at which nitrogen from the second source enters the river was 80.063 hundred pounds per year.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

For any $c > 0$, the presence of $c\int_0^{40} B(t)\, dt$ will earn the first point.

8-3

A copy error in the integrand imported from part (c) will earn the first point. Any new copy error will not earn the first point. In either case the second point can be earned if the correct answer is given.

Read for correctness or consistency with any positive value of $\int_0^{40} A(t)\, dt$ imported from part (a).

The second point is not earned for a correct value with no supporting work.

The second point is earned for an answer that is one of the four values $8368, 8369, 8368.35$, or 8368.351 (with supporting work).

If one of the values $8368, 8369, 8368.35$, or 8368.351 is presented but then rounded to 8400 (e.g., $8368 \approx 8400$), the response will earn the second point. However, a response of just 8400, without one of these four values present, will not earn the second point.

Units are not scored in this part, except an answer of $836{,}800$ (or $836{,}900$) must clearly be labeled as *pounds* in order to earn the second point.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ $\int_0^{40} B(t)\, dt$
- ☐ Answer

Solution:

$$\text{Total} = \int_0^{40} A(t)\, dt + \int_0^{40} B(t)\, dt \approx 4982 + 3386.350701 \approx 8368$$

The total combined amount of nitrogen from the two sources that enters the river over the time interval $0 \leq t \leq 40$ is approximately 8368 hundred pounds.

8-3

49. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Diego runs back and forth along a straight track. During the time interval $0 \le t \le 30$ seconds, Diego's velocity, in feet per second, is modeled by the function v given by $v(t) = \frac{200 \sin\left(\frac{t^2}{75}\right)}{t+1}$.

(a) What is the first time, t_1, that Diego changes direction? Find Diego's average velocity over the time interval $0 \le t \le t_1$.

(b) At time $t = 30$, how far is Diego from his position at time $t = 0$?

(c) Find the total distance that Diego runs during the 30 seconds from time $t = 0$ to time $t = 30$.

(d) Let $a(t)$ represent Diego's acceleration during the time interval $0 < t < 30$ seconds. Using correct units, explain the meaning of $\int_5^{10} a(t)\, dt$ in the context of the problem. Find the value of $\int_5^{10} a(t)\, dt$.

Part A

For the first point, supporting work stating that $v(t)$ changes sign is required.

For the second point, supporting work that includes the average value expression with an appropriate definite integral is required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

$t_1 = 15.350$

8-3

- ☐
- ☐ average value integral
- ☐ answer

Solution:

Diego changes direction when $v(t)$ changes sign. The first time this happens is at $t_1 = 15.350$ (or 15.349).

$$\frac{1}{t_1 - 0}\int_0^{t_1} v(t)\, dt = 10.503965$$

Diego's average velocity over the time interval $0 \le t \le t_1$ is 10.504 (or 10.503) feet per second.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ definite integral
- ☐ answer

Solution:

$$\int_0^{30} v(t)\, dt = 128.322555$$

At time $t = 30$, Diego is 128.323 (or 128.322) feet from his position at time $t = 0$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ definite integral

8-3

- [ ] answer

Solution:

$$\int_0^{30} |v(t)|\, dt = 243.425185$$

During the 30 seconds from time $t = 0$ to time $t = 30$, the total distance that Diego runs is 243.425 feet.

Part D

The first point requires 3 things: (1) net change in Diego's velocity, (2) units of feet per second, and (3) reference to time interval from $t = 5$ to $t = 10$.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] interpretation with units
- [ ] answer

Solution:

Since acceleration is the rate of change of velocity, $\int_5^{10} a(t)\, dt$ is the net change in Diego's velocity, in feet per second, during the 5 seconds from time $t = 5$ to time $t = 10$.

$$\int_5^{10} a(t)\, dt = \int_5^{10} v'(t)\, dt = v(10) - v(5) = 6.765 \text{ feet per second}$$

8-3

50. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Gwen runs back and forth along a straight track. During the time interval $0 \leq t \leq 45$ seconds, Gwen's velocity, in feet per second, is modeled by the function v given by $v(t) = \frac{250 \sin\left(\frac{t^2}{120}\right)}{t+6}$.

(a) What is the first time, t_1, that Gwen changes direction? Find Gwen's average velocity over the time interval $0 \leq t \leq t_1$.

(b) At time $t = 45$, how far is Gwen from her position at time $t = 0$?

(c) Find the total distance that Gwen runs during the 45 seconds from time $t = 0$ to time $t = 45$.

(d) Let $a(t)$ represent Gwen's acceleration during the time interval $0 < t < 45$ seconds. Find the value of $\int_{15}^{30} a(t)\, dt$. Using correct units, explain the meaning of $\int_{15}^{30} a(t)\, dt$ in the context of the problem.

Part A

For the first point, supporting work stating that $v(t)$ changes sign is required.

For the second point, supporting work that includes the average value expression with an appropriate definite integral is required. To be eligible for the second point, the presented value of t_1 may be incorrect but must be positive.

The third point requires the first two points to have been earned.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3 ✓

8-3

The student response accurately includes all three of the criteria below.

- ☐ $t_1 = 19.416$
- ☐ average value integral
- ☐ answer

Solution:

Gwen changes direction when $v(t)$ changes sign. The first time this happens is at $t_1 = 19.416$.

$$\frac{1}{t_1 - 0}\int_0^{t_1} v(t)\,dt = 7.264377$$

Gwen's average velocity over the time interval $0 \le t \le t_1$ is 7.264 feet per second.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ definite integral
- ☐ answer

Solution:

$$\int_0^{45} v(t)\,dt = 116.537972$$

At time $t = 45$, Gwen is 116.538 (or 116.537) feet from her position at time $t = 0$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

8-3

- [ ] definite integral
- [ ] answer

Solution:

$$\int_{0}^{45} |v(t)|\ dt = 250.099483$$

During the 45 seconds from time $t = 0$ to time $t = 45$, the total distance that Gwen runs is 250.099 feet.

Part D

The second point requires 3 things: (1) net change in Gwen's velocity, (2) units of feet per second, and (3) reference to time interval from $t = 15$ to $t = 30$.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] answer
- [ ] interpretation with units

Solution:

$$\int_{15}^{30} a(t)\ dt = \int_{15}^{30} v'(t)\ dt = v(30) - v(15) = -4.844 \text{ feet per second}$$

Since acceleration is the rate of change of velocity, $\int_{15}^{30} a(t)\ dt$ is the net change in Gwen's velocity, in feet per second, during the 15 seconds from time $t = 15$ to time $t = 30$.

8-3

51. The penguin population on an island is modeled by a differentiable function P of time t, where $P(t)$ is the number of penguins and t is measured in years, for $0 \le t \le 40$. There are 100,000 penguins on the island at time $t = 0$. The birth rate for the penguins on the island is modeled by

$$B(t) = 1000e^{0.06t} \text{ penguins per year}$$

and the death rate for the penguins on the island is modeled by

$$D(t) = 250e^{0.1t} \text{ penguins per year.}$$

(a) What is the rate of change of the penguin population on the island at time $t = 0$?

(b) To the nearest whole number, what is the penguin population on the island at time $t = 40$?

(c) To the nearest whole number, what is the average rate of change of the penguin population on the island for $0 \le t \le 40$?

(d) To the nearest whole number, find the absolute minimum penguin population and the absolute maximum penguin population on the island for $0 \le t \le 40$. Show the analysis that leads to your answer.

Part A

The response can earn up to 1 point:

1 point is earned for the answer

$P'(0) = B(0) - D(0) = 1000 - 250 = 750$ penguins per year

0	1

The response earns all of the following point:

1 point is earned for the answer

$P'(0) = B(0) - D(0) = 1000 - 250 = 750$ penguins per year

Part B

The response can earn up to 2 points:

1 point is earned for correct limits

1 point is earned for the integrand

1 point is earned for answer

$$P(40) = 100000 + \int_0^{40} (B(t) - D(t))\, dt$$

$= 100000 + 33057.56459$

8-3

There are 133,058 penguins on the island.

0	1	2	3

The student earns all of the following point:

1 point is earned for correct limits

1 point is earned for the integrand

1 point is earned for answer

$$P(40) = 100000 + \int_0^{40} (B(t) - D(t))\, dt$$

$$= 100000 + 33057.56459$$

There are 133,058 penguins on the island.

Part C

The response can earn up to 1 point:

1 point is earned for the answer

$$\frac{1}{40} \int_0^{40} (B(t) - D(t))\, dt = 826.439$$

OR

$$\frac{P(40) - P(0)}{40 - 0} = \frac{133058 - 100000}{40} = 826.45$$

The average rate of change is 826 penguins per year.

0	1

The student earns all of the following point:

1 point is earned for the answer

$$\frac{1}{40} \int_0^{40} (B(t) - D(t))\, dt = 826.439$$

OR

8-3

$$\frac{P(40)-P(0)}{40-0}=\frac{133058-100000}{40}=826.45$$

The average rate of change is 826 penguins per year.

Part D

The response can earn up to 4 points:

1 point is earned for the expression $B(t)$ - $D(t) = 0$

$B(t)$ - $D(t) = 0$

1 point is earned for the solution for t

$$1000e^{0.06t}=250e^{0.1t}\Rightarrow t=A=\frac{\ln 4}{0.04}=34.657359$$

The absolute minimum and absolute maximum occur at a critical point or at an endpoint.

1 point is earned for the minimum value

1 point is earned for the maximum value

$P(0) = 100000$

$$P(A) = 100000+\int_0^A (B(t)-D(t))\,dt = 139166.667$$

$P(40) = 133058$

The minimum population is 100,000 and the maximum population is 139,167 penguins.

0	1	2	3	4

The student earns all of the following point:

1 point is earned for the expression $B(t)$ - $D(t) = 0$

$B(t)$ - $D(t) = 0$

1 point is earned for the solution for t

$$1000e^{0.06t}=250e^{0.1t}\Rightarrow t=A=\frac{\ln 4}{0.04}=34.657359$$

The absolute minimum and absolute maximum occur at a critical point or at an endpoint.

1 point is earned for the minimum value

8-3

1 point is earned for the maximum value

$P(0) = 100000$

$P(A) = 100000 + \int_0^A (B(t) - D(t))\, dt = 139166.667$

$P(40) = 133058$

The minimum population is 100,000 and the maximum population is 139,167 penguins.

52. If $\frac{dy}{dt} = 6e^{-0.08(t-5)^2}$, by how much does y change as t changes from $t = 1$ to $t = 6$?

(A) 3.870

(B) 8.341

(C) 18.017

(D) 22.583

Answer D

Correct. The change in y from $t = 1$ to $t = 6$ is $y(6) - y(1)$. By the Fundamental Theorem of Calculus, $y(6) - y(1) = \int_1^6 y'(t)dt = \int_1^6 6e^{-0.08(t-5)^2} dt = 22.583$, where the numerical integration is done with the calculator.

8-3

53. In a small town, water is being used at a rate of $r(t) = 12 + 9\sin(0.5t - 1.6)$ thousand gallons per hour, where t is measured in hours since midnight. To the nearest integer, how much water, in thousands of gallons, is used in the town during the 6-hour period from 3 A.M. ($t = 3$) to 9 A.M. ($t = 9$) ?

(A) 3

(B) 76

(C) 107

(D) 125

Answer C

Correct. The total change in the amount of water used in the town from $t = 3$ to $t = 9$ is given by the definite integral of the rate of change of the amount of water used.

$$\int_3^9 r(t)\,dt = \int_3^9 (12 + 9\sin(0.5t - 1.6))\,dt = 107.387$$

To the nearest integer, the town used 107 thousands of gallons of water.

54. A particle moves along a line so that its position at any time $t \geq 0$ is given by $s(t)$. If $s''(t) = 2 - \sqrt{t^3 + 1}$ and $s'(1) = 1.5$, then the velocity of the particle at $t = 2.5$ is

(A) positive and increasing

(B) positive and decreasing

(C) negative and increasing

(D) negative and decreasing

Answer B

Correct. The velocity of the particle at $t = 2.5$ is

$$v(2.5) = v(1) + \int_1^{2.5} v'(t)dt = s'(1) + \int_1^{2.5} s''(t)dt = 1.5 + \int_1^{2.5} \left(2 - \sqrt{t^3 + 1}\right)dt = 0.607.$$

Therefore, the velocity is positive at $t = 2.5$. In addition, $v'(2.5) = s''(2.5) = -2.077 < 0$, so the velocity is decreasing at $t = 2.5$.

8-3

55. At time $t = 0$ hours, snow begins to fall. The rate at which snow accumulates is given by $s\left(t\right) = t^{\frac{3}{2}} \cdot e^{-\frac{t}{2}}$ inches per hour. How many inches of snow accumulate during the time interval $3 \le t \le 6$ hours?

(A) 0.428

(B) 0.987

(C) 2.256

(D) 2.961

Answer D

Correct. The total change in snow accumulation during the time interval $3 \le t \le 6$ hours is given by the definite integral of the rate of snow accumulation during that time interval.

$$\int_3^6 s(t)\,dt = \int_3^6 t^{\frac{3}{2}} \cdot e^{-\frac{t}{2}}\,dt = 2.961$$

The definite integral is evaluated with the calculator.

8-3

56. The rate at which raw sewage enters a treatment tank is given by $E(t) = 850 + 715\cos\left(\frac{\pi t^2}{9}\right)$ gallons per hour for $0 \le t \le 4$ hours. Treated sewage is removed from the tank at the constant rate of 645 gallons per hour. The treatment tank is empty at time $t = 0$.

(a) How many gallons of sewage enter the treatment tank during the time interval $0 \le t \le 4$? Round your answer to the nearest gallon.

(b) For $0 \le t \le 4$, at what time t is the amount of sewage in the treatment tank greatest? To the nearest gallon, what is the maximum amount of sewage in the tank? Justify your answers.

(c) For $0 \le t \le 4$, the cost of treating the raw sewage that enters the tank at time t is $(0.15 - 0.02t)$ dollars per gallon. To the nearest dollar, what is the total cost of treating all the sewage that enters the tank during the time interval $0 \le t \le 4$?

Part A

1 point is earned for integral

1 point is earned for the answer

$$\int_0^4 E(t)\,dt = 3981 \text{ gallons}$$

0	1	2

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

$$\int_0^4 E(t)\,dt = 3981 \text{ gallons}$$

Part B

1 point is earned if sets $E(t) = 645$

1 point is earned if identifies $t = 2.309$ as a candidate

1 point is earned for the amount of sewage at $t = 2.309$

1 point is earned for the conclusion

8-3

Let $S(t)$ be the amount of sewage in the treatment tank at time t. Then $S'(t) = E(t) - 645$ and $S'(t) = 0$ when $E(t) = 645$. On the interval $0 \le t \le 4$, $E(t) = 645$ when $t = 2.309$ and $t = 3.559$.

t (hours)	amount of sewage in treatment tank
0	0
2.309	$\int_0^{2.309} E(t)\,dt - 645(2.309) = 1637.178$
3.559	$\int_0^{3.559} E(t)\,dt - 645(3.559) = 1228.520$
4	$3981.022 - 645(4) = 1401.022$

The amount of sewage in the treatment tank is greatest at $t = 2.309$ hours. At that time, the amount of sewage in the tank, rounded to the nearest gallon, is 1637 gallons.

0	1	2	3	4

The student earns all of the following points:

1 point is earned if sets $E(t) = 645$

1 point is earned if identifies $t = 2.309$ as a candidate

1 point is earned for the amount of sewage at $t = 2.309$

1 point is earned for the conclusion

Let $S(t)$ be the amount of sewage in the treatment tank at time t. Then $S'(t) = E(t) - 645$ and $S'(t) = 0$ when $E(t) = 645$. On the interval $0 \le t \le 4$, $E(t) = 645$ when $t = 2.309$ and $t = 3.559$.

t (hours)	amount of sewage in treatment tank
0	0
2.309	$\int_0^{2.309} E(t)\,dt - 645(2.309) = 1637.178$
3.559	$\int_0^{3.559} E(t)\,dt - 645(3.559) = 1228.520$
4	$3981.022 - 645(4) = 1401.022$

The amount of sewage in the treatment tank is greatest at $t = 2.309$ hours. At that time, the amount of sewage in the tank, rounded to the nearest gallon, is 1637 gallons.

8-3

Part C

1 point is earned for integrand

1 point is earned for limits

1 point is earned for the answer

Total cost = $\int_0^4 (0.15 - 0.02t)\, dt = 474.320$

The total cost of treating the sewage that enters the tank during the time interval $0 \le t \le 4$ to the nearest dollar, is \$474.

0	1	2	3

The student earns all of the following points:

1 point is earned for integrand

1 point is earned for limits

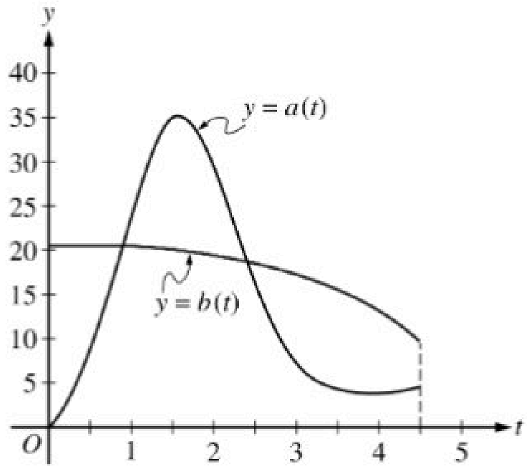
1 point is earned for the answer

Total cost = $\int_0^4 (0.15 - 0.02t)\, dt = 474.320$

The total cost of treating the sewage that enters the tank during the time interval $0 \le t \le 4$ to the nearest dollar, is \$474.

8-3

57.



During the time interval $0 \leq t \leq 4.5$ hours, water flows into tank *A* at a rate of $a\left(t\right)=\left(2t-5\right)+5e^{2\sin t}$ liters per hour. During the same time interval, water flows into tank *B* at a rate of $b(t)$ liters per hour. Both tanks are empty at time $t = 0$. The graphs of $y = a(t)$ and $y = b(t)$, shown in the figure above, intersect at $t = k$ and $t = 2.416$.

(a) How much water will be in tank *A* at time $t = 4.5$?

(b) During the time interval $0 \leq t \leq k$ hours, water flows into tank *B* at a constant rate of 20.5 liters per hour. What is the difference between the amount of water in tank *A* and the amount of water in tank *B* at time $t = k$?

(c) The area of the region bounded by the graphs of $y = a(t)$ and $y = b(t)$ for $k \leq t \leq 2.416$ is 14.470. How much water is in tank *B* at time $t = 2.416$?

(d) During the time interval $2.7 \leq t \leq 4.5$ hours, the rate at which water flows into tank *B* is modeled by $w\left(t\right)=21-\frac{30t}{\left(t-8\right)^2}$ liters per hour. Is the difference $w(t)$ - a(t) increasing or decreasing at time $t = 3.5$? Show the work that leads to your answer.

Part A

1 point is earned for integral

1 point is earned for answer

$$\int_0^{4.5} a\left(t\right)dt=66.532128$$

At time $t = 4.5$, tank *A* contains 66.532 liters of water.

0	1	2

8-3

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for answer

$$\int_0^{4.5} a(t)dt = 66.532128$$

At time $t = 4.5$, tank A contains 66.532 liters of water.

Part B

1 point is earned for sets $a(k) = 20.5$

1 point is earned for integral

1 point is earned for answer

$$a(k) = 20.5 \Rightarrow k = 0.892040$$
$$\int_0^{k} (20.5 - a(t))dt = 10.599191$$

At time $t = k,$ the difference in the amounts of water in the tanks is 10.599 liters.

0	1	2	3

The student response earns all of the following points:

1 point is earned for sets $a(k) = 20.5$

1 point is earned for integral

1 point is earned for answer

$$a(k) = 20.5 \Rightarrow k = 0.892040$$
$$\int_0^{k} (20.5 - a(t))dt = 10.599191$$

At time $t = k,$ the difference in the amounts of water in the tanks is 10.599 liters.

Part C

8-3

1 point is earned for $\int_{k}^{2.416} a(t)dt$

1 point is earned for answer

$$\int_{0}^{2.416} b(t)dt = \int_{0}^{k} b(t)dt + \int_{k}^{2.416} b(t)dt$$
$$\int_{0}^{k} b(t)dt = 20.5 \cdot k = 18.286826$$

On $k < t < 2.416$, tank A receives $\int_{k}^{2.416} a(t)dt = 44.497051$ liters of water, which is 14.470 more liters of water than tank B.

Therefore, $\int_{k}^{2.416} b(t)dt = \int_{k}^{2.416} a(t)dt - 14.470 = 30.027051.$

$$\int_{0}^{k} b(t)dt + \int_{k}^{2.416} b(t)dt = 48.313876$$

At time $t = 2.416$, tank B contains 48.314 (or 48.313) liters of water.

0	1	2

The student response earns all of the following points:

1 point is earned for $\int_{k}^{2.416} a(t)dt$

1 point is earned for answer

$$\int_{0}^{2.416} b(t)dt = \int_{0}^{k} b(t)dt + \int_{k}^{2.416} b(t)dt$$
$$\int_{0}^{k} b(t)dt = 20.5 \cdot k = 18.286826$$

8-3

On $k < t < 2.416$, tank A receives $\int_k^{2.416} a(t)dt = 44.497051$ liters of water, which is 14.470 more liters of water than tank B.

Therefore, $\int_k^{2.416} b(t)dt = \int_k^{2.416} a(t)dt - 14.470 = 30.027051.$

$$\int_0^k b(t)dt + \int_k^{2.416} b(t)dt = 48.313876$$

At time $t = 2.416$, tank B contains 48.314 (or 48.313) liters of water.

Part D

1 point is earned for $w'(3.5) - a'(3.5) < 0$

1 point is earned for conclusion

$w'(3.5) - a'(3.5) = -1.14298 < 0$

The difference $w(t) - a(t)$ is decreasing at $t = 3.5$.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for $w'(3.5) - a'(3.5) < 0$

1 point is earned for conclusion

$w'(3.5) - a'(3.5) = -1.14298 < 0$

The difference $w(t) - a(t)$ is decreasing at $t = 3.5$.

8-3

58.

t (days)	0	10	22	30
$W'(t)$ (GL per day)	0.6	0.7	1.0	0.5

The twice-differentiable function W models the volume of water in a reservoir at time t, where $W(t)$ is measured in gigaliters (GL) and t is measured in days. The table above gives values of $W'(t)$ sampled at various times during the time interval $0 \le t \le 30$ days. At time $t = 30$, the reservoir contains 125 gigaliters of water.

(a) Use the tangent line approximation to W at time $t = 30$ to predict the volume of water $W(t)$, in gigaliters, in the reservoir at time $t = 32$. Show the computations that lead to your answer.

(b) Use a left Riemann sum, with the three subintervals indicated by the data in the table, to approximate $\int_0^{30} W'(t)dt$. Use this approximation to estimate the volume of water $W(t)$, in gigaliters, in the reservoir at time $t = 0$. Show the computations that lead to your answer.

(c) Explain why there must be at least one time t, other than t = 10, such that W'(t) = 0.7 GL/day.

(d) The equation $A = 0.3W^{2/3}$ gives the relationship between the area A, in square kilometers, of the surface of the reservoir, and the volume of water $W(t)$, in gigaliters, in the reservoir. Find the instantaneous rate of change of A, in square kilometers per day, with respect to t when $t = 30$ days.

Part A

The response can earn up to 1 point:

1 point: For the correct answer

An equation of the tangent line is $y = 0.5(t - 30) + 125$.

$W(32) \approx 0.5(32 - 30) + 125 = 126$

0	1

The response earns all of the following point:

1 point: For the correct answer

An equation of the tangent line is $y = 0.5(t - 30) + 125$.

$W(32) \approx 0.5(32 - 30) + 125 = 126$

Part B

8-3

The response can earn up to 3 points:

1 point: For the left Riemann sum

1 point: For the approximation

1 point: For the correct answer

$$\int_0^{30} W'(t)dt \approx (10)(0.6) + (12)(0.7) + (8)(1.0) = 22.4$$

$$W(0) = W(30) - \int_0^{30} W'(t)dt = 125 - 22.4 = 102.6$$

0	1	2	3

The response earns all of the following points:

1 point: For the left Riemann sum

1 point: For the approximation

1 point: For the correct answer

$$\int_0^{30} W'(t)dt \approx (10)(0.6) + (12)(0.7) + (8)(1.0) = 22.4$$

$$W(0) = W(30) - \int_0^{30} W'(t)dt = 125 - 22.4 = 102.6$$

Part C

The response can earn up to 2 points:

2 points: For the explanation

W' is differentiable $\Rightarrow$ W' is continuous.

$W'(30) = 0.5 < 0.7 < 1.0 = W'(22)$

By the Intermediate Value Theorem, there must be at least one time t, $22 \le t \le 30$, such that $W'(t) = 0.7$.

8-3

0	1	2

The response earns all of the following points:

2 points: For the explanation

W' is differentiable ⇒ W' is continuous.

$W'(30) = 0.5 < 0.7 < 1.0 = W'(22)$

By the Intermediate Value Theorem, there must be at least one time t, $22 \le t \le 30$, such that $W'(t) = 0.7$.

Part D

The response can earn up to 3 points:

2 points: For stating $\dfrac{dA}{dt}$

1 point: For the correct answer

$$\frac{dA}{dt} = (0.3)\frac{2}{3}W^{-1/3} \cdot \frac{dW}{dt} = \frac{0.2}{\sqrt[3]{W}} \cdot \frac{dW}{dt}$$
$$\left.\frac{dA}{dt}\right|_{t=30} = \frac{0.2}{\sqrt[3]{125}} \cdot 0.5 = 0.02$$

0	1	2	3

The response earns all of the following points:

2 points: For stating $\dfrac{dA}{dt}$

1 point: For the correct answer

$$\frac{dA}{dt} = (0.3)\frac{2}{3}W^{-1/3} \cdot \frac{dW}{dt} = \frac{0.2}{\sqrt[3]{W}} \cdot \frac{dW}{dt}$$
$$\left.\frac{dA}{dt}\right|_{t=30} = \frac{0.2}{\sqrt[3]{125}} \cdot 0.5 = 0.02$$